

MCAK Recognizes the Nucleotide-Dependent Feature at Growing Microtubule Ends

Wei Chen, Yin-Long Song, Jian-Feng He, Xin Liang 

IDG/McGovern Institute for Brain Research, State Key Laboratory of Complex, Severe, and Rare Diseases, School of Life Sciences, Tsinghua University, Beijing, China

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eLife Assessment

This work presents **valuable** new information on the microtubule-binding mode of the microtubule kinesin-13, MCAK, the authors use quantitative single-molecule studies to propose that MCAK preferentially binds to a GDP-Pi-tubulin portion of the microtubule end. However, the evidence provided to support this claim remains **incomplete** and would benefit from more rigorous methodology particularly the diffraction limited experiments do not provide sufficient spatial resolution to support the authors' conclusions. In addition, a more through discussion of the existing literature would further strengthen the manuscript.

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Abstract

The growing plus-end is a key regulatory site for microtubule dynamics. MCAK (mitotic centromere-associated kinesin), a microtubule depolymerizing kinesin, is an end-binding regulator of catastrophe frequency. It is intriguing how MCAK specifically binds to growing microtubule ends. Here, we measure the end-binding kinetics of MCAK using single-molecule imaging, and reveal that MCAK binds to the entire GTP cap, including the region where EB1 preferentially binds and the distalmost tip. Further analysis shows that MCAK strongly binds to GTPγS microtubules, and this binding preference is dependent on the nucleotide state of MCAK. This finding suggests that MCAK recognizes the nucleotide-dependent feature of microtubules. Moreover, we show that although MCAK and XMAP215 partly share binding regions at the distalmost ends, their regulatory effects can be additive, highlighting a simple rule of how the end-binding regulators coordinate their activities. Overall, our findings provide new insights into how MCAK regulates microtubule end dynamics.

Introduction

Microtubules are essential components of the eukaryotic cytoskeleton, supporting a wide range of cellular processes, including cell division, polarization, and migration (Alberts, 2015). However, their properties can vary considerably across different cellular contexts and structures (Janke and Magiera, 2020; Kapitein and Hoogenraad, 2015; Wittmann et al., 2001). This variability reflects the complex and tightly regulated mechanisms that govern the dynamics of cellular microtubules, among which the microtubule-associated proteins (MAPs) play crucial roles.

Kinesin superfamily members not only facilitate the transport of cargo along microtubules but also serve as regulators of microtubule dynamics, as exemplified by the kinesin-13 family. Initially identified through studies of spindle functions and neurons (Aizawa et al., 1992; Noda et al., 1995; Walczak et al., 1996; Wordeman and Mitchison, 1995), kinesin-13 has since been implicated in a broad range of cellular processes, including spindle assembly (Walczak et al., 2002), chromosome segregation (Kline-Smith et al., 2004), directional migration (Zong et al., 2021), ciliary length control (Piao et al., 2009; Vasudevan et al., 2015), and neuronal polarization (Ghosh-Roy et al., 2012; Puri et al., 2021). The loss of kinesin-13 often results in altered structural and dynamic properties of cellular microtubules. Moreover, the subcellular localizations of kinesin-13, such as at growing microtubule ends, centrosomes, and centromeres, are key regulatory sites for microtubule dynamics. These observations suggest that kinesin-13 family members are key regulators for the length, mass, and polarity of cellular microtubules.

As a representative member of the kinesin-13 family, MCAK has been characterized as both a microtubule depolymerase and a catastrophe factor (Desai et al., 1999; Gardner et al., 2011; Hunter et al., 2003; Kline-Smith and Walczak, 2002; Walczak et al., 1996). It has been proposed that MCAK targets growing microtubule ends by interacting with end-binding proteins (EBs) or motors, such as Kif18b (Gouveia et al., 2010; Honnappa et al., 2009; McHugh and Welburn, 2023; Tanenbaum et al., 2011). Upon binding, MCAK destabilizes the protective GTP cap at the growing microtubule ends by bending and disassembling the underlying substrate, including terminal tubulin dimers or protofilaments (Benoit et al., 2018; Moores and Milligan, 2006; Wang et al., 2017), thereby increasing the frequency of catastrophe events. MCAK can also function independently (Desai et al., 1999; Gardner et al., 2011; Hunter et al., 2003), potentially targeting stabilized microtubule ends through direct binding or rapid diffusion along the microtubule lattice (Desai et al., 1999; Helenius et al., 2006; Wang et al., 2012). Previous structural studies suggest that MCAK recognizes curved protofilaments at microtubule ends (Asenjo et al., 2013; Benoit et al., 2018; Tan et al., 2008; Trofimova et al., 2018; Wang et al., 2017). However, these studies primarily focus on stabilized microtubules, short protofilaments, or tubulin dimers, leaving the mechanism by which MCAK binds to growing microtubule ends unresolved.

In this study, we use single-molecule imaging to investigate the direct binding of MCAK to growing microtubule ends. A key finding is that MCAK binds throughout the entire GTP cap, encompassing both the EB cap, where EB1 preferentially binds, and the distalmost cap, where XMAP215 binds to curved protofilaments. Notably, this is the first discovery that MCAK exhibits a binding preference to the EB cap. Further analysis reveals that MCAK strongly binds to GTP γ S microtubules, which are thought to mimic GDP-Pi-tubulin conformations enriched in the EB cap. Having confirmed that MCAK binds to the entire GTP cap structure, we next examine how MCAK co-regulates microtubule dynamics in collaboration with XMAP215, and reveal their additive, rather than mutually exclusive, regulatory effects. Overall, this study provides novel insights into how MCAK binds to and acts at growing microtubule ends.

Results

MCAK shows a binding preference for growing microtubule ends

We began by measuring the single-molecule binding kinetics of MCAK at growing microtubule ends using the in vitro microtubule dynamics assay. This experiment was performed in BRB80 supplemented with 50 mM KCl and 1 mM ATP, providing a near-physiological ion strength of approximately 0.2 M, similar to what was previously reported (Scopes, 2002; Storey, 2004). The size-exclusion chromatography analysis showed that the purified GFP-MCAK exists as dimers under this condition (Fig. s1).

We observed MCAK binding events both at growing microtubule ends and along the lattice (Fig. 1A). To characterize these events, we measured the fluorescence intensity of each binding event (see Methods) and plot a probability distribution of all binding events (Fig. s1). We used the range of fluorescence intensity from the probability distribution ($\mu \pm 2\sigma$: 300 ± 145 A.U.) to identify single MCAK dimers. Using a custom software (Maurer et al., 2014; Song et al., 2020), we determined the end-binding position of each individual MCAK molecule in each frame by fitting a Gaussian function to the intensity profile along the longitudinal axis of the microtubule, with the plus-end serving as the positional reference (Fig. 1B). The average position over the dwell time of each MCAK molecule was taken as the position of this binding event (Fig. s2). From this, we generated a probability distribution of all binding positions, derived from 152 binding events across three assays (Fig. 1C). This analysis revealed that the MCAK binding region was 162 nm in length (FWHM, Full Width at Half Maximum), corresponding to approximately 20 layers of tubulin dimers. We categorized the binding events of MCAK within this region as “end binding” and those located more proximally as “lattice binding”.

To investigate the binding kinetics, we first compared the apparent association rates of MCAK on the plus-end ($k_{\text{on-P}}$) and the lattice ($k_{\text{on-L}}$) in the presence of ATP. The ratio ($R_{\text{E/L}}$) of $k_{\text{on-P}}$ ($(22.6 \pm 1.4) \times 10^{-6} \text{ s}^{-1} \text{ nM}^{-1}$, per tubulin dimer, mean $\pm$ SEM, hereinafter unless otherwise specified) to $k_{\text{on-L}}$ ($(2.8 \pm 0.3) \times 10^{-6} \text{ s}^{-1} \text{ nM}^{-1}$) was 13 ± 1 ($n=66$ microtubules from 3 assays) (Fig. 1D-E). We also measured the dwell time, which reflects the dissociation rate (k_{off}) (Fig. s1). MCAK exhibited a longer dwell time at the plus-end than on the lattice (plus end: 0.75 ± 0.02 s, $n=966$ events from 3 assays; lattice: 0.62 ± 0.02 s, $n=702$ events from 3 assays, $p<0.001$) (Fig. 1F). Using these kinetic parameters, we calculated the dissociation constant (K_{d}) of MCAK for individual tubulin dimers at the plus-end as $69 \mu\text{M}$. This affinity may appear relatively low, but when considering the microtubule end as a multivalent receptor with multiple MCAK-binding sites (end length=160 nm) - where only a few bound MCAK molecules are required to regulate microtubule dynamics parameters - the effective dissociation constant reaches the nanomolar range (260 nM). The MCAK's lattice binding affinity was even lower ($K_{\text{d-L}}=1057 \mu\text{M}$, per tubulin dimer), demonstrating that MCAK has a clear end-binding preference. We also observed single-molecule binding events of MCAK at the minus-ends (Fig. 1A), with a similar $R_{\text{E/L}}$ (13 ± 2 , $n=28$ microtubules from 3 assays) and K_{d} ($40 \mu\text{M}$, per tubulin dimer). We focused on the plus-end binding events (referred to as “end-binding” hereinafter), due to their greater physiological relevance.

Having measured the single-molecular binding events, we next investigated whether the binding of MCAK to growing microtubule ends occurs through direct end-binding (3D diffusion) or 1D lattice-diffusion in our experimental conditions (Gouveia et al., 2010; Helenius et al., 2006). These two mechanisms are not mutually exclusive. Our data revealed that the lattice-diffusion coefficient of MCAK (D) on the lattice of dynamic microtubules was $0.023 \pm 0.001 \mu\text{m}^2 \cdot \text{s}^{-1}$ (Fig. 1G), comparable to the values reported under high-salt conditions (Cooper et al., 2010; Gouveia et al., 2010), but lower than that in low-salt conditions (Helenius et al., 2006). Because the dwell time of the lattice-binding events was approximately 0.6 s, the average length scanned by

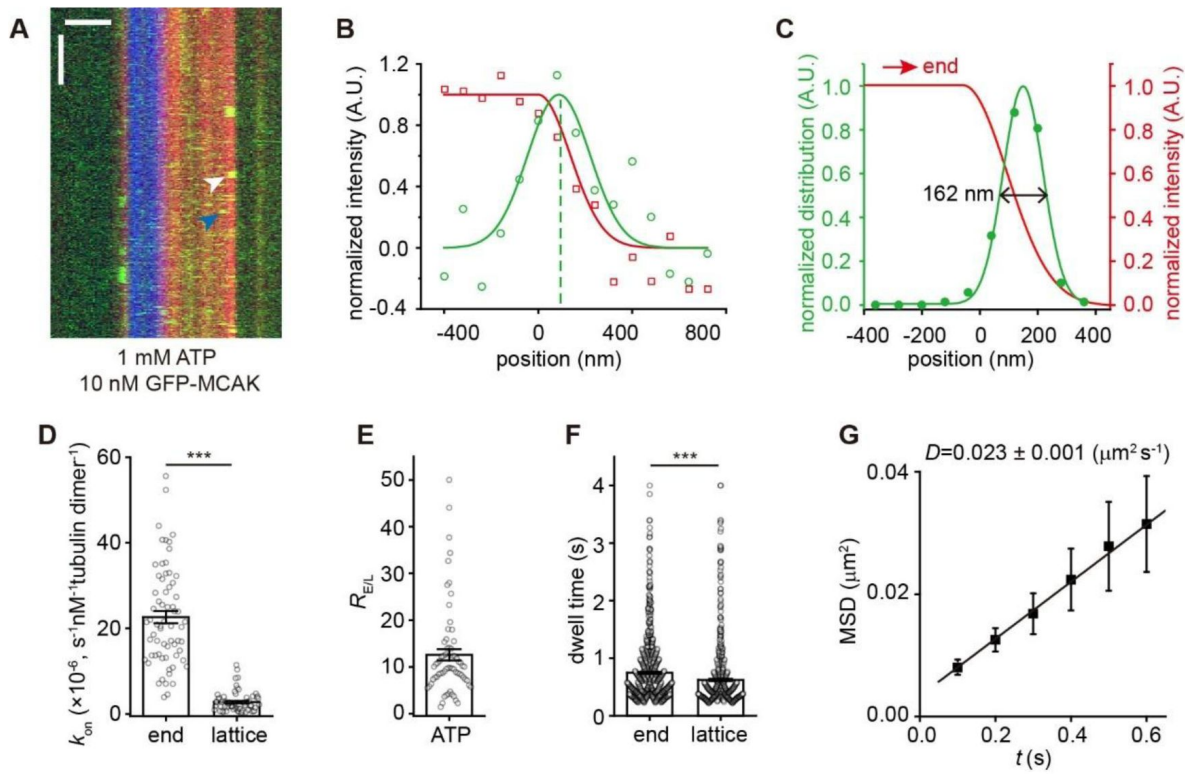


Figure 1.

MCAK preferentially binds to growing microtubule ends

(A) Representative kymographs of single-molecule GFP-MCAK (green, 10 nM, measured in terms of monomer concentration unless otherwise stated) binding events on dynamic microtubules (red, tubulin: 16 mM) growing from GMPCPP microtubule seeds (blue) in the presence of 1 mM ATP. A plus-end binding event and a lattice binding event were marked by the white and blue arrowheads, respectively. Scale bars: vertical, 5 s; horizontal, 2 μm .

(B) Representative intensity profiles of a single GFP-MCAK molecule (green, 10 nM) and a growing microtubule end (red, tubulin: 16 mM) at one frame during the dwell time of a binding event (corresponding to the white arrowhead in panel A). The peak position of a GFP-MCAK molecule was determined by Gaussian fitting (green dashed line).

(C) Position probability distributions of GFP-MCAK binding sites (green, n=152 events) along the microtubule longitudinal axis. The red curve represents the fitting curve of the growing microtubule end as the positional reference. The MCAK binding region was defined as the area within the FWHM (double black arrow, 162 nm) of the fitted green curve.

(D) Statistical quantification of on-rate (k_{on}) of GFP-MCAK's binding to the plus-end versus lattice of dynamic microtubules in the presence of 1 mM ATP (n=66 microtubules from 3 assays).

(E) Statistical quantification of $R_{\text{E/L}}$ in the presence of 1 mM ATP (n=66 microtubules from 3 assays).

(F) Statistical quantification of the dwell time of GFP-MCAK's binding to the growing end versus lattice of dynamic microtubules in the presence of 1 mM ATP (n=966 events from 3 assays for the plus end; n=702 events from 3 assays for lattice).

(G) Mean-squared displacement (MSD) of GFP-MCAK plotted against the time interval (t , 0.1s per frame). The diffusion coefficient (D) was calculated via linear regression of the equation ($\langle x^2 \rangle = 2Dt$), yielding $0.023 \mu\text{m}^2 \text{s}^{-1}$ (error bars represent SEM, n=203 trajectories).

In panel D, E and F, all the data were presented as mean $\pm$ SEM. All comparisons were performed using two-tailed Mann-Whitney U test with Bonferroni correction, n.s., no significance; ***, $p < 0.001$.

a MCAK molecule via 1D lattice-diffusion was ~ 160 nm, roughly the length the GTP cap. Using an established method (Helenius et al., 2006 [DOI](#)), we calculated the flux of MCAK arriving at growing ends via 1D diffusion was 0.004 s^{-1} . This was significantly lower than the observed MCAK arrival rate at the plus-end observed ($N_p = k_{on} \cdot C = 0.067 \text{ s}^{-1}$. C : MCAK concentration). Therefore, in our experimental conditions, 1D lattice-diffusion contributes minimally to the growing end-binding of MCAK ($\sim 6\%$), suggesting that the end-binding primarily results from the direct binding pathway.

MCAK binds to the entire GTP cap

We sought to understand how MCAK recognizes growing microtubule ends. Previous studies, mostly based on tubulin dimers, short protofilaments, or stabilized microtubules, suggesting that MCAK binds microtubule ends by recognizing curved protofilaments (Asenjo et al., 2013 [DOI](#); Benoit et al., 2018 [DOI](#); Trofimova et al., 2018 [DOI](#)). This model predicts that MCAK binds to the distalmost end of growing microtubules, overlapping with the binding region of XMAP215 family members (i.e. the distalmost cap) (Ayaz et al., 2012 [DOI](#); Brouhard et al., 2008 [DOI](#)). To test this hypothesis, we quantified the end-binding regions of MCAK, EB1, and XMAP215 (**Fig. 2A-B** [DOI](#)) by measuring the FWHM of their spatial distribution. To our surprise, the probability distribution of MCAK's positions exhibited a significant difference from that of XMAP215 ($p < 10^{-5}$, Kolmogorov-Smirnov test). The end-binding region of MCAK (FWHM=185 nm) was longer than that of XMAP215 (FWHM=123 nm) (**Fig. 2B** [DOI](#)). It extended proximally towards the lattice, covering the binding region of EB1 (i.e. the EB cap, FWHM=118 nm). To further understand whether MCAK binds to the distalmost cap and the EB cap with different kinetics, we compared the dwell time of MCAK at the distal and proximal locations (**Fig. 2B** [DOI](#)). As shown in **Fig. 2B** [DOI](#), MCAK molecules located to the left of the XMAP215 binding region were classified as proximally distributed, while those located to the right of the EB1 binding region were considered distally distributed. No significant difference was found between the two groups ($\tau_{\text{distal}} = 0.8 \pm 0.1 \text{ s}$, $n=28$ binding events; $\tau_{\text{proximal}} = 0.7 \pm 0.1 \text{ s}$, $n=45$ binding events; $p>0.3$, the two-tailed Mann-Whitney U test with Bonferroni correction). Additionally, no significant differences were observed in the intensity profiles of the microtubule ends in the presence of MCAK (1 nM), EB1 (10 nM) and XMAP215 (1 nM), respectively ($\text{End}_{\text{EB1}} \text{ vs } \text{End}_{\text{MCAK}}$, $p=0.68$; $\text{End}_{\text{MCAK}} \text{ vs } \text{End}_{\text{XMAP215}}$, $p=0.68$; $\text{End}_{\text{EB1}} \text{ vs } \text{End}_{\text{XMAP215}}$, $p=1$. Kolmogorov-Smirnov test) (**Fig. 2C** [DOI](#)).

The results suggest that MCAK's end-binding region partially overlaps with that of XMAP215, with more MCAK positioned behind XMAP215, closer to the lattice. To confirm this, we assessed the colocalization of MCAK and XMAP215. Given the low probability of both molecules simultaneously appearing at microtubule ends at low concentrations, we increased the concentration of XMAP215 to ensure its consistent tracking of growing microtubule ends while simultaneously recording single-molecule binding events of MCAK. In each frame, we measured the distance between the intensity peaks of XMAP215 and MCAK along the microtubule's longitudinal axis (**Fig. 2D** [DOI](#)). The average distance over MCAK's binding duration was then calculated to determine its relative position to XMAP215. Subsequently, we plotted a probability distribution of all relative distances (**Fig. 2E** [DOI](#)). This distribution peaked at -65 nm, indicating that MCAK predominantly binds behind XMAP215, with 14.5% of binding events occurring within XMAP215's binding region (distance > 0). These findings support the notion that MCAK binds to both the distalmost cap and the EB cap.

The end-binding preference of MCAK is nucleotide state-dependent

It has been reported that MCAK tightly binds to microtubules in both the AMPPNP (a non-hydrolysable ATP analogue, mimicking the ATP-bound state) and nucleotide-free states, but exhibits weaker binding in the ADP-bound state (Friel and Howard, 2011 [DOI](#); Helenius et al., 2006 [DOI](#); Wang et al., 2012 [DOI](#)). Moreover, kinesin-13s in the AMPPNP state preferentially form protofilament curls or rings with tubulins (Asenjo et al., 2013 [DOI](#); Benoit et al., 2018 [DOI](#); Tan et al., 2008 [DOI](#)). These findings suggest that the interaction between kinesin-13s and microtubules is regulated by their nucleotide-bound state. Given this, we sought to determine whether the end-binding preference of

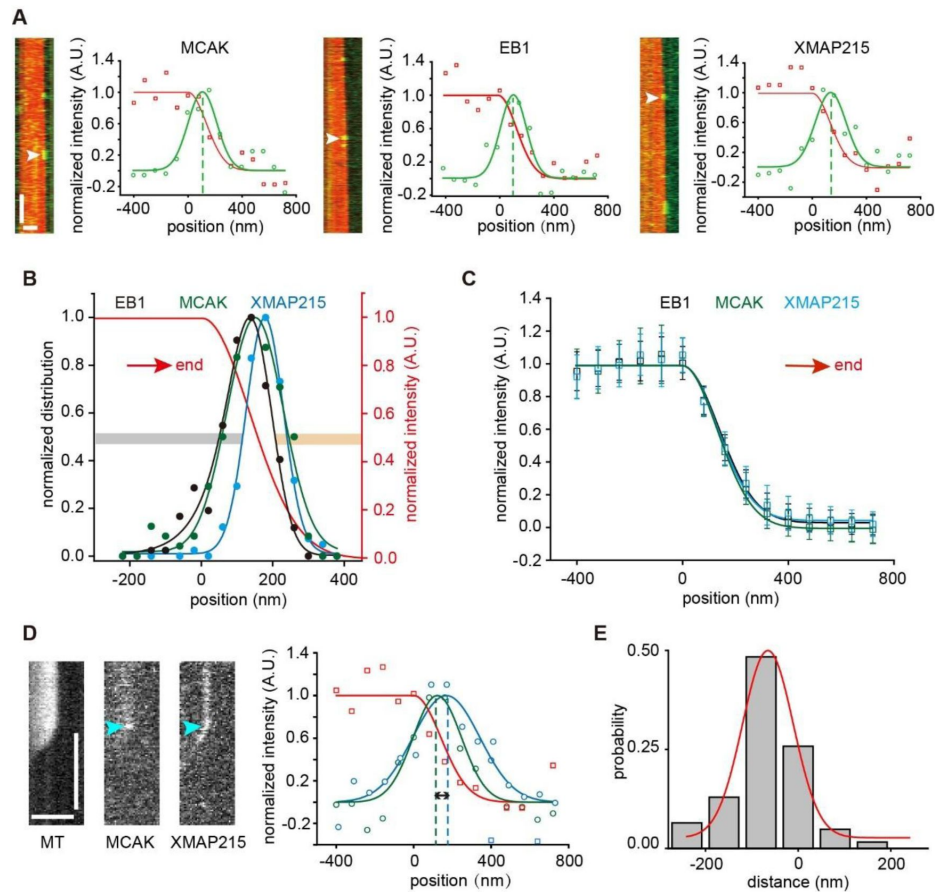


Figure 2.

MCAK binds to the entire GTP cap of growing microtubule ends

(A) Representative kymographs showing the individual binding events of GFP-MCAK (1 nM, with 1 mM ATP, left), EB1-GFP (10 nM, middle) or XMAP215-GFP (1 nM, right) at growing microtubule ends (red, tubulin: 12 μ M). The plots showed the representative intensity profiles of single molecules (green) and growing microtubule ends (red) at one frame during the dwell time of the individual binding event (white arrowheads). Peak positions were determined by Gaussian fitting (green dashed line). Scale bars: vertical, 5 s; horizontal, 2 μ m.

(B) Position probability distributions of the binding sites of GFP-MCAK, EB1-GFP (black, $n=184$ events) and XMAP215-GFP (blue, 142 events) along the microtubule longitudinal axis. The fitting curve of the growing microtubule end (red) was used as the positional reference. The binding region was defined as the area within the FWHM of the fitted curves. The GFP-MCAK molecules localized to the left of XMAP215-GFP's binding region was defined as proximally distributed (grey bar), and those localized to the right of EB1-GFP's binding region was defined as distally distributed (orange bar).

(C) Averaged intensity profiles of the growing microtubule ends decorated with GFP-MCAK (green, 123 microtubules), EB1-GFP (black, 184 microtubules) and XMAP215-GFP (blue, 142 microtubules). Data were presented as mean $\pm$ std, and no significant difference was observed between the three conditions (Kolmogorov-Smirnov test, End_{EB1} vs End_{MCAK} , $p=0.68$; End_{MCAK} vs $\text{End}_{\text{XMAP215}}$, $p=0.68$; End_{EB1} vs $\text{End}_{\text{XMAP215}}$, $p=1$). In panels **B** and **C**, the red arrows indicated microtubule growth direction.

(D) Representative kymographs showing concurrent binding events of XMAP215-GFP (25 nM) and MCAK-RFP (20 nM) at growing microtubule ends (tubulin: 12 μ M) in the presence of 1 mM ATP. The plots showed the representative intensity profiles of XMAP215-GFP (blue), MCAK-RFP (green) and growing microtubule ends (red) at one frame during the dwell time of the binding event (cyan arrowhead). Peak positions were determined by Gaussian fitting (dashed line). The spacing between the two peaks represents the instantaneous distance between MCAK and XMAP215 (double black arrow). Scale bars: vertical, 5 s; horizontal, 2 μ m.

(E) Probability distribution of the distance between XMAP215-GFP and MCAK-RFP (62 events from 3 assays) at growing microtubule ends. In this plot, the position of XMAP215-GFP was taken as the origin. A negative distance indicated a more proximal localization.

MCAK is also nucleotide state-dependent. To explore this, we measured the end-binding kinetics of MCAK in complex with AMPPNP, ADP, or in the nucleotide-free (APO) state (**Fig. 3A**), under conditions that maintain similar microtubule growth rates (**Fig. s1**).

Comparing the binding kinetics of MCAK in different nucleotide states, we found that $k_{\text{on-P}}$ and $k_{\text{on-L}}$ of MCAK·AMPPNP were both lower than those of MCAK·ATP (**Fig. 3 B-C**), yet resulting in a similar end-binding preference ($R_{\text{E/L}}=11 \pm 2$, $n=29$ microtubules from 2 assays, $p>1$). However, the binding of MCAK·AMPPNP at the end and lattice both showed a longer dwell time compared to that of MCAK·ATP (3.32 ± 0.24 s at the end, $n=231$ events from 2 assays, $p<0.001$; 1.46 ± 0.21 s at lattice, $n=142$ events from 2 assays, $p<0.001$), indicating lower k_{off} (**Fig. 3D** and **Fig. s1**). As a result of the proportionally reduced $k_{\text{on-P}}$ and k_{off} , the end-binding affinity of MCAK·AMPPNP ($K_{\text{d-P}}=61$ μM , per tubulin dimer, hereinafter) was comparable to that of MCAK·ATP. In contrast, MCAK·ADP showed nearly unchanged $k_{\text{on-P}}$ but increased $k_{\text{on-L}}$, leading to a lower $R_{\text{E/L}}$ (2 ± 0.3 , $n=21$ microtubules from 2 assays, $p<0.001$) and suggesting the reduction, but not a complete loss, of the end-binding preference (**Fig. 3 B-C**). Additionally, MCAK·ADP showed a significantly shorter end-binding and lattice-binding dwell time (0.46 ± 0.02 s at the end, $n=327$ events from 2 assays, $p<0.001$) compared to MCAK·ATP (**Fig. 3 D**). Consequently, MCAK·ADP has a lower end-binding affinity ($K_{\text{d-P}}=163$ μM). MCAK·APO, the nucleotide-free state, showed a large increase in $k_{\text{on-P}}$ and $k_{\text{on-L}}$, thereby having a lower $R_{\text{E/L}}$ (2 ± 0.2 , $n=36$ microtubules from 3 assays, $p<0.001$) (**Fig. 3 B-C**). It also showed a significantly longer end-binding dwell time (1.89 ± 0.23 s, $n=71$ events from 3 assays, $p<0.001$) (**Fig. 3D**), indicating a lower k_{off} (**Fig. s1**). As a result, MCAK·APO had a higher end-binding affinity ($K_{\text{d-P}}=1.8$ μM), but it partially loses the end-binding preference. Overall, these findings indicate that MCAK's end-binding preference and affinity are dependent on its nucleotide state, with the ATP-bound state exhibiting the highest end-binding preference.

MCAK strongly binds to GTPyS microtubules

The binding region of MCAK at growing microtubule ends encompasses the EB cap. This region consists mostly of GDP·Pi-tubulin dimers, which are thought to form at least some lateral contacts and adopt a straighter conformation (Guesdon et al., 2016; Maurer et al., 2012). This raises the question of how MCAK recognizes the EB cap. To understand this, we examined that to what extent, MCAK may adopt the end-binding mechanism utilized by EB1.

We first investigated the contribution of the nucleotide state-dependent feature of the EB cap, which underlies the end-binding preference of EBs (Maurer et al., 2011; Maurer et al., 2012). Specifically, we compared the binding affinities of MCAK·ATP on GTPyS and GDP microtubules, mimicking the tubulin dimers in the EB cap and the lattice of growing microtubules, respectively (**Fig. 4A**). Because MCAK·ATP at the concentration required for full lattice decoration would rapidly disassemble microtubules, we cannot directly compare the total binding of MCAK on microtubules as what was previously done for the EBs (Maurer et al., 2011; Maurer et al., 2012). Instead, we adopted an alternative approach by recording the binding of MCAK·ATP at a low concentration (10 nM) on microtubules and measuring fluorescence intensity summation over a period of time (1000 frames, 0.3s per frame) (**Fig. s3**). Strikingly, quantitative analysis revealed that MCAK·ATP showed a significantly stronger binding to GTPyS microtubules than to GDP microtubules (**Fig. 4 B-C**). Because the tubulin dimers in GTPyS microtubules were considered to be the analogue of GDP·Pi-tubulins in the EB cap (Manka and Moores, 2018; Maurer et al., 2011; Maurer et al., 2014; Zhang et al., 2015), this result provides a biochemical basis for the MCAK·ATP's binding preference for the EB cap. Here, we used EB1 as a control (**Fig. s4**).

Given that the growing end-binding affinity and preference of MCAK depend on the nucleotide state, we wondered whether its preference for GTPyS microtubules follows a similar trend. To address this, we performed the experiments for MCAK·AMPPNP, MCAK·ADP and MCAK·APO. The MCAK variants bound to the lattice of microtubules with different affinities (MCAK·APO>MCAK·AMPPNP>MCAK·ADP > MCAK·ATP) (**Fig. 4B**), consistent with the results observed on the lattice of dynamic microtubules. Additionally, MCAK maintained a binding preference for GTPyS

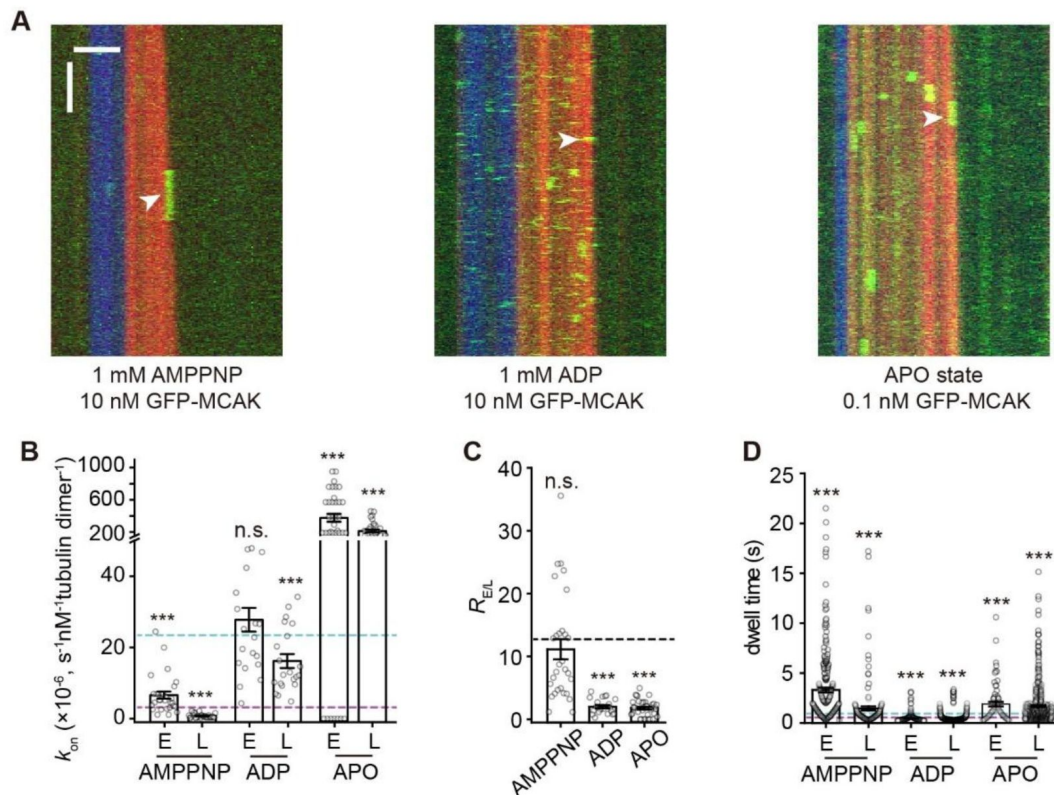


Figure 3.

The growing end-binding preference of MCAK is dependent on its nucleotide state

(A) Representative kymographs of single-molecule GFP-MCAK (green) binding events on dynamic microtubules (red, tubulin: 16 μ M) growing from the GMPCPP microtubule seeds (blue) in the presence of 1 mM AMPPNP (left), 1 mM ADP (middle) and APO (right) (the nucleotide-free state). The plus-end binding events were indicated by white arrowheads. Scale bar: vertical, 5 s; horizontal, 2 μ m.

(B) Statistical quantification of on-rate (k_{on}) of GFP-MCAK binding to the plus end (E) versus the lattice (L) of dynamic microtubules in the presence of 1 mM AMPPNP (n=29 microtubules from 2 assays), 1 mM ADP (n=21 microtubules from 2 assays) and APO (n=36 microtubules from 3 assays). The cyan and purple dashed lines denoted the on-rate value of MCAK binding at the plus end and lattice in the ATP condition (from Fig. 1), respectively.

(C) Statistical quantification of $R_{E/L}$ in the presence of 1 mM AMPPNP (n=29 microtubules from 2 assays), 1 mM ADP (n=21 microtubules from 2 assays) and APO (n=36 microtubules from 3 assays). Dashed line: the mean value of $R_{E/L}$ in the ATP condition from Fig. 1.

(D) Statistical quantification of the dwell time for GFP-MCAK binding at the growing ends and lattice of dynamic microtubules in the presence of 1 mM AMPPNP (n=231 events from 2 assays for the plus end; n=142 events from 2 assays for lattice), 1 mM ADP (n=327 events from 2 assays for the plus end; n=1184 events from 2 assays for lattice) and APO (n=71 events from 3 assays for the plus end; n=573 events from 3 assays for lattice). The cyan and purple dashed lines denoted the dwell time value of MCAK binding at the plus end and lattice in the ATP condition (from Fig. 1), respectively.

In the panel B, C and D, all the data were presented as mean $\pm$ SEM. All the comparisons were made against the corresponding value in the ATP condition from Fig. 1 using two-tailed Mann-Whitney U test with Bonferroni correction, n.s., no significance; ***, p < 0.001.

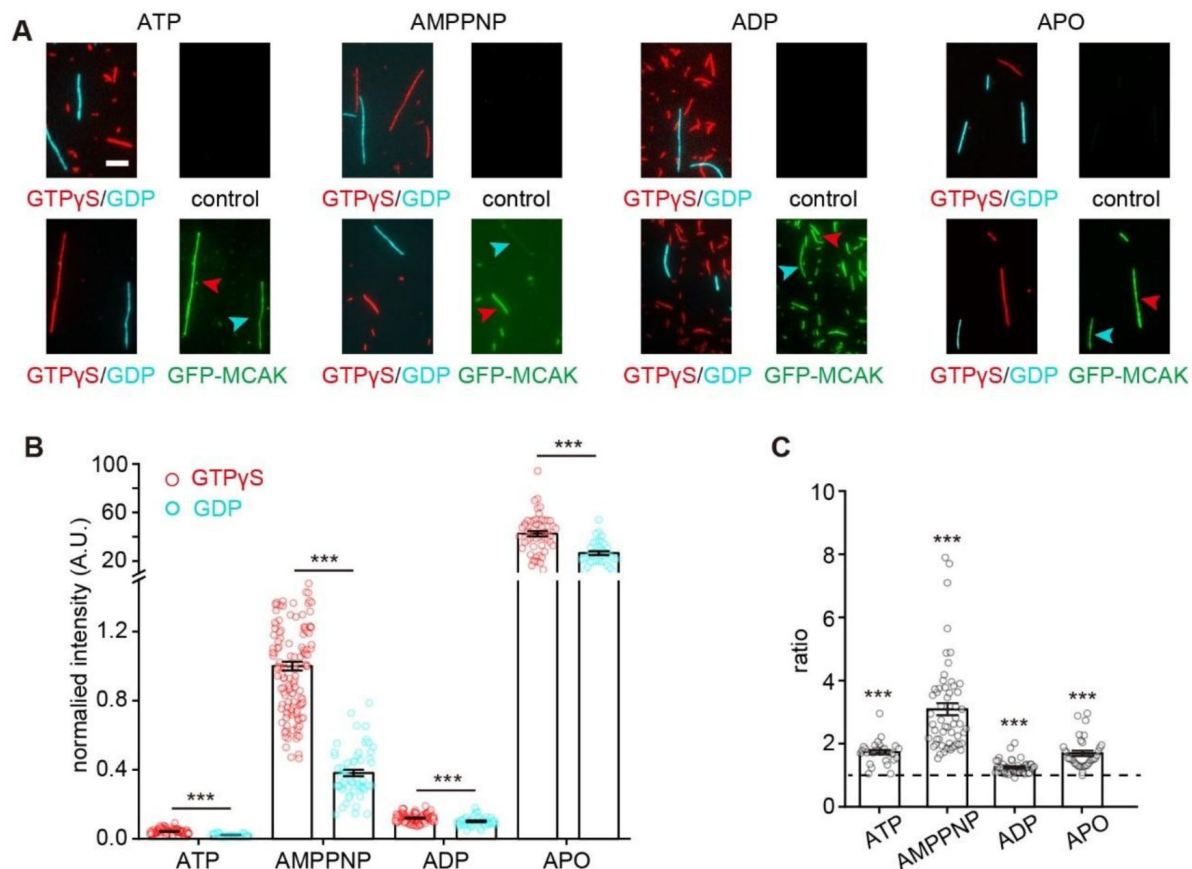


Figure 4.

MCAK strongly binds to GTPyS microtubules in a nucleotide-dependent manner

(A) Representative projection images of GFP-MCAK binding to GTPyS microtubules (red) versus GDP microtubules (cyan) in the presence of 1 mM ATP (left, 10 nM GFP-MCAK), 1 mM AMPPNP (left middle, 1 nM GFP-MCAK), 1 mM ADP (right middle, 10 nM GFP-MCAK) and at the APO state (right, 0.1 nM GFP-MCAK). Note that the binding on the GDP microtubules was compared to that on the GTPyS microtubules in the same flow cell. The GFP-MCAK binding to the GTPyS and GDP microtubules was indicated by the red and cyan arrowhead, respectively. Scale bar: 5 μ m.

(B) Statistical quantification of the normalized fluorescence intensity of GFP-MCAK (1 nM) on different microtubules in the presence of 1 mM ATP (95 GTPyS microtubules, 28 GDP microtubules from 2 assays), 1 mM AMPPNP (133 GTPyS microtubules, 56 GDP microtubules from 3 assays), 1 mM ADP (111 GTPyS microtubules, 44 GDP microtubules from 2 assays) or at the APO state (52 GTPyS microtubules, 36 GDP microtubules from 3 assays). All data were normalized to the binding to GTPyS microtubules binding in the AMPPNP condition.

(C) The binding intensity ratios of GFP-MCAK at GTPyS microtubules to GDP microtubules under various nucleotide conditions. ATP: n=28, from 2 assays. AMPPNP: n=56 GTPyS/GDP from 3 assays. ADP: n=44 from 2 assays. APO: n=36 from 3 assays in APO state. The dashed line represented 1. Statistical comparisons were performed between the ratios and 1. In panels **B** and **C**, all the data were presented as mean $\pm$ SEM. All comparisons were performed using two-tailed Mann-Whitney U test with Bonferroni correction, ***, $p < 0.001$.

microtubules over GDP microtubules across all nucleotide states, but the degree of binding preference varied with the nucleotide state ($\text{MCAK} \cdot \text{AMPPNP} > \text{MCAK} \cdot \text{ATP} \approx \text{MCAK} \cdot \text{APO} > \text{MCAK} \cdot \text{ADP}$) (**Fig. 4C**). This finding indicates that MCAK's binding preference for GTPyS microtubules (i.e., GDP-Pi-tubulin) is strongest at the ATP state, mirroring its preference for growing microtubule ends.

MCAK binds to the end region of stabilized GMPCPP microtubules in a nucleotide state-dependent manner

Our findings also support the notion that MCAK binds to the distalmost cap, consistent with the previous reports that MCAK binds to the curved protofilaments of microtubule ends (Asenjo et al., 2013; Benoit et al., 2018; Desai et al., 1999; Tan et al., 2008; Wang et al., 2017). Given the nucleotide state-dependence of the growing end-binding preference, we questioned whether this binding preference for curved protofilaments also depends on the nucleotide state of MCAK. To investigate this issue, we analyzed the binding of MCAK at the ends of stabilized GMPCPP microtubules (**Fig. 5A**), where curved protofilaments are thought to resemble those at the ends of dynamic microtubules, but without the presence of GDP-Pi-tubulins (Atherton et al., 2018; Manka and Moores, 2018; McIntosh et al., 2018). Taxol stabilized GDP-microtubules were here used as a control (**Fig. s5**). We found that MCAK, in both the ATP and AMPPNP state, exhibited a binding preference for the ends of GMPCPP-stabilized microtubules (**Fig. 5A-F**). Similarly, MCAK-ATP showed a binding preference for the ends of taxol-stabilized GDP microtubules, albeit to a lesser extent (**Fig. s5**). This observation suggests that the ends of stable microtubules mimic the structure of growing microtubule ends to varying degrees, with the GMPCPP-stabilized ends providing the closest resemblance. In contrast, MCAK-APO and MCAK-ADP, did not show the end-binding preference (**Fig. 5A-F**), consistent with the observation that MCAK-ADP and MCAK-APO also had a lower binding preference for growing microtubule ends.

The $\alpha 4$ helix and the L2 loop of MCAK contribute to the binding preference to the EB cap

Intuitively, the binding preference for the EB cap, in addition to the previously established preference for the curved protofilaments, provides more binding sites for MCAK at growing microtubule ends. We sought to identify the key domains responsible for MCAK's preference for the EB cap. Previous studies have shown that the $\alpha 4$ helix and the L2 loop are critical for mediating MCAK's interaction with curved protofilaments (Asenjo et al., 2013; Benoit et al., 2018; Patel et al., 2016; Tan et al., 2008; Wang et al., 2017). Specifically, the $\alpha 4$ helix of MCAK interacts with the intradimer interface of the tubulin heterodimer, while the L2 loop contacts the interdimer interface (**Fig. 6A**). Moreover, the electrostatic interactions between the $\alpha 4$ helix and the L2 loop of MCAK with tubulin heterodimer are important for MCAK's interactions with microtubule ends (Patel et al., 2016; Tan et al., 2008), as phosphorylation at T537 disrupts MCAK's microtubule-end binding and depolymerization activity (Sanhaji et al., 2010). We then wondered whether these domains also contribute to the binding preference for the EB cap. To test this hypothesis, we introduced two mutations: the K524A mutation in the $\alpha 4$ helix, which is expected to disrupt the electrostatic interaction between MCAK and the tubulin heterodimer (Patel et al., 2016; Wang et al., 2017), and the V298S mutation in the L2 loop, which is predicted to impair the hydrophobic interaction between the L2 loop and α -tubulin (Wang et al., 2015). Consistent with previously reports (Wang et al., 2017; Wang et al., 2015), both mutants exhibited deficiencies in depolymerization and reduced ATPase activity (**Fig. s6**).

Compared to the wild-type MCAK, both $\text{MCAK}^{\text{K524A}}$ and $\text{MCAK}^{\text{V298S}}$ were capable of binding to growing microtubule ends (**Fig. 6B**), but their $k_{\text{on-P}}$ and dwell time at growing microtubule ends were significantly reduced (**Fig. 6C-D**). This indicates that the binding affinity of the mutants to growing microtubule ends is substantially decreased. Here, due to the minimal to negligible

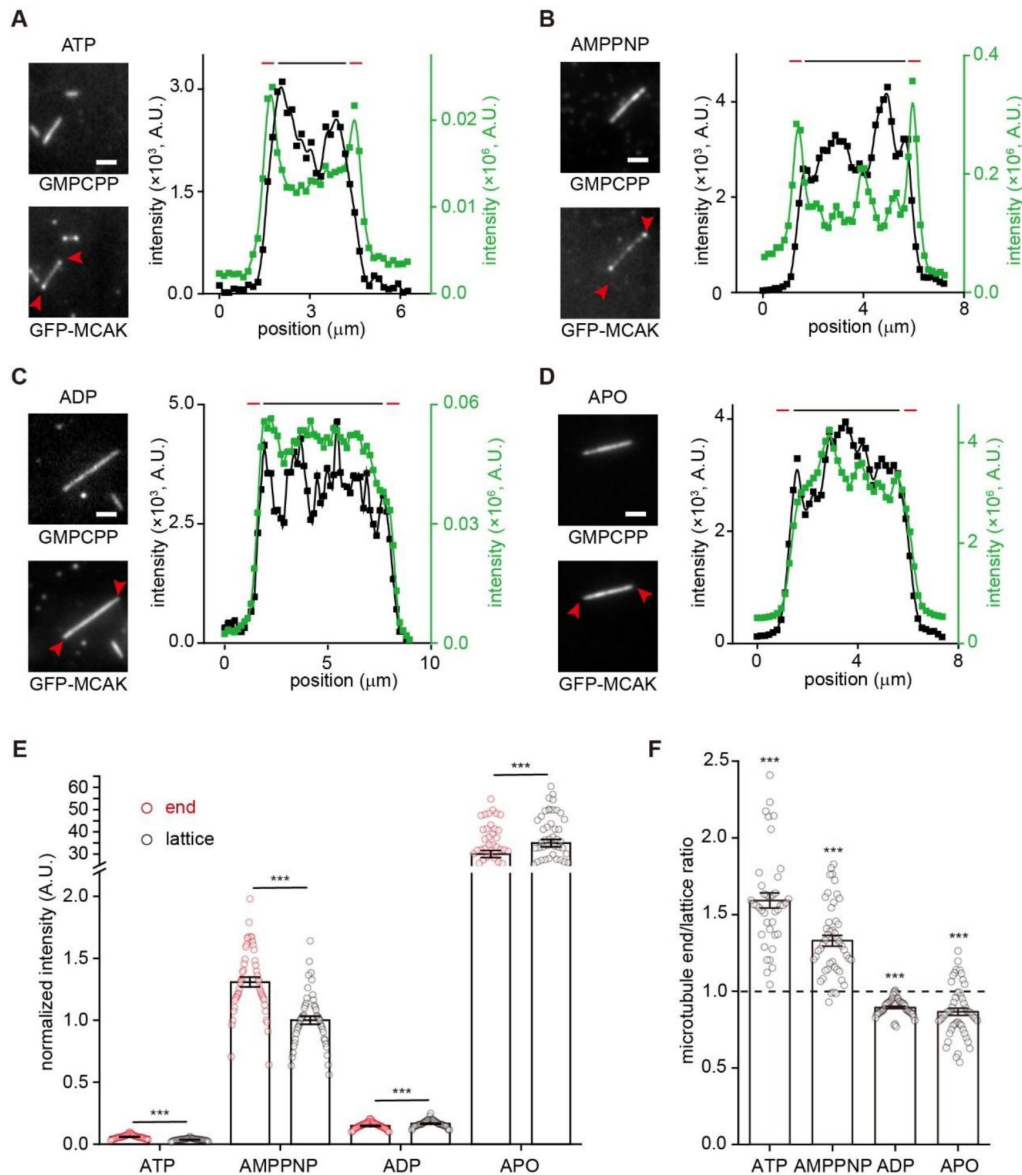


Figure 5.

MCAK binds to the ends of GMPCPP microtubules in a nucleotide state-dependent manner

(A, B, C, D) Representative fluorescence projection images and intensity profiles showing GFP-MCAK binding to the ends and lattice of GMPCPP microtubules in the presence of 1 mM ATP (10 nM GFP-MCAK), 1 mM AMPPNP (1 nM GFP-MCAK), 1 mM ADP (10 nM GFP-MCAK) or at the APO state (0.1 nM GFP-MCAK). Red arrowhead: the signal of MCAK enriching at the ends. Intensity profiles (right panels) showed spatial intensity signal distributions of the microtubule (black curve) and GFP-MCAK (green curve). Red and black bars indicated the end and the lattice, respectively. Scale bar = 2 μm .

(E) Statistical quantification of the normalized binding intensity of GFP-MCAK (1 nM) at the end versus lattice regions of GMPCPP microtubules in the presence of 1 mM ATP (40 microtubules from 3 assays), 1 mM AMPPNP (47 microtubules from 3 assays), 1 mM ADP (45 microtubules from 3 assays) or at the APO state (52 microtubules from 3 assays). All the data were normalized to the binding intensity of GFP-MCAK at GMPCPP microtubule lattices in the AMPPNP condition. All the data were presented as mean $\pm$ SEM. Statistical analysis was performed using two-tailed paired *t*-test by Bonferroni correction, ***, $p < 0.001$.

(F) The end-to-lattice binding intensity ratios of GFP-MCAK at GMPCPP microtubules under various nucleotide conditions. The ratios were presented as mean $\pm$ SEM. The dashed line represents 1. Statistical comparisons were performed between the ratios and 1 using two-tailed Mann-Whitney U test with Bonferroni correction, ***, $p < 0.001$.

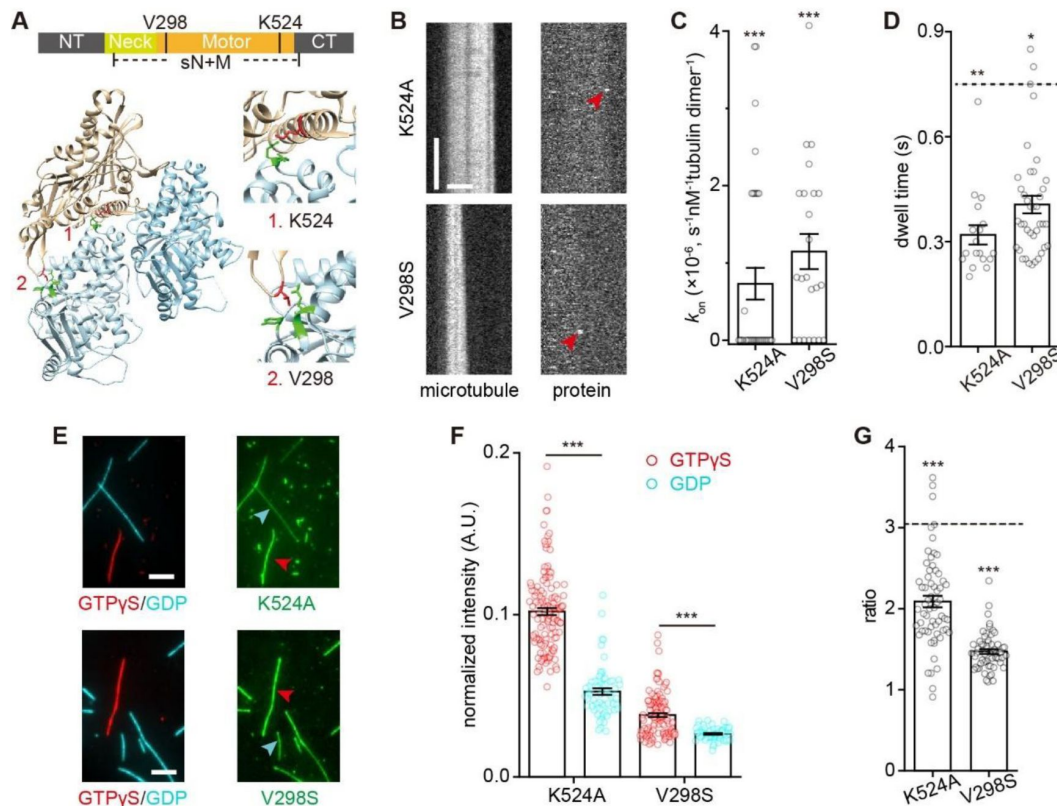


Figure 6.

The $\alpha 4$ helix and the L2 loop of MCAK contribute to the preferential binding to the EB cap

(A) Structural model of the MCAK^{SN+M}-tubulin complex (PDB ID: 5MIO; X-ray diffraction). Top: Domain organization of MCAK. Bottom: Enlarged view of $\alpha 4$ helix and L2 loop, regions mediating the interaction between MCAK and tubulin, with the key sites (K524 and V298) were highlighted in red. The residues in tubulin that potentially interact with K524 and V298 were highlighted in green.

(B) Representative kymographs of single-molecule binding events of GFP-MCAK^{K524A} (10 nM) and GFP-MCAK^{V298S} (30 nM) at growing microtubules (red, tubulin: 16 μ M) in the presence of 1 mM ATP. The end-binding events were indicated by red arrowheads. Scale bars: vertical, 5 s; horizontal, 2 μ m.

(C) Statistical quantification of on-rate (k_{on-P}) of GFP-MCAK^{K524A} (31 microtubules from 3 assays) and GFP-MCAK^{V298S} (23 microtubules from 2 assays) at growing microtubule ends in the presence of 1 mM ATP. The data were presented as mean $\pm$ SEM and were compared to that of wild-type GFP-MCAK (from Fig. 1).

(D) Statistical quantification of dwell time of GFP-MCAK^{K524A} (18 binding events from 3 assays) and GFP-MCAK^{V298S} (39 binding events from 3 assays) at growing microtubule ends in the presence of 1 mM ATP. The data were presented as mean $\pm$ SEM and were compared to that of wild-type GFP-MCAK in the presence of ATP (from Fig. 1). The dashed line: the mean dwell time of wild-type GFP-MCAK.

(E) Representative fluorescence projection images of 5 nM GFP-MCAK^{K524A} and 10 nM GFP-MCAK^{V298S} binding to GTP γ S (red arrowhead) versus GDP (cyan arrowhead) microtubules in the presence of 1 mM AMPPNP. Scale bar: 5 μ m.

(F) Statistical quantification of normalized binding intensity of 1 nM GFP-MCAK^{K524A} (144 GTP γ S microtubules and 62 GDP microtubules from 3 assays) and GFP-MCAK^{V298S} (124 GTP γ S microtubules, and 59 GDP microtubules from 3 assays) on different microtubules. All the data were normalized to the binding intensity of wild-type GFP-MCAK at GTP γ S microtubules in the AMPPNP condition (from Fig. 4). Data were presented as mean $\pm$ SEM.

(G) The binding intensity ratios of GFP-MCAK^{K524A} or GFP-MCAK^{V298S} on GTP γ S microtubules to that on GDP microtubules in the presence of 1 mM AMPPNP. GFP-MCAK^{K524A}: n=62 from 3 assays in AMPPNP state. GFP-MCAK^{V298S}: n=59 GTP γ S/GDP binding ratios from 3 assays in AMPPNP state. Dashed line: the GTP γ S/GDP ratio of wild-type GFP-MCAK. The data were presented as mean $\pm$ SEM and were compared to that for wild-type GFP-MCAK (from Fig. 4).

In panel C, D, F and G, all comparisons were performed using two-tailed Mann-Whitney U test with Bonferroni correction, n.s., no significance; *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$.

number of observed binding events of the mutants at growing microtubule ends and the lattice, reliable calculations of $R_{E/L}$ and k_{off} were not feasible for the mutants. Therefore, we directly assayed the binding preference for GTPγS microtubules mimicking the EB cap. Both mutants showed significantly reduced binding intensity and a decreased, but not the loss of, preference for GTPγS microtubules compared to wild-type MCAK (**Fig. 6E-G** vs **Fig. 4B-C**). These results suggest that both the α4 helix and the L2 loop contributes the binding preference for the EB cap.

Functional specification of MCAK and XMAP215

MCAK is a microtubule catastrophe factor, whereas XMAP215 functions as a microtubule polymerase (Brouhard et al., 2008; Desai et al., 1999; Gardner et al., 2011). Both are capable of binding to dynamic microtubule ends and regulating microtubule dynamics. In the previous reports, they were shown to antagonistically regulate microtubule assembly (Barr and Gergely, 2008; Kinoshita et al., 2001; Moriwaki and Goshima, 2016; Tournebise et al., 2000). Additionally, structural models have indicated that both MCAK and XMAP215 bind to curved tubulin dimers (Ayaz et al., 2012; Trofimova et al., 2018), raising the possibility of competition for binding sites in the distalmost cap. However, while XMAP215 was reported to promote microtubule catastrophe (Farmer et al., 2021), our findings showed that MCAK binds more proximally relative to XMAP215 at growing microtubule ends. The extent to which competition from XMAP215 would antagonize the function of MCAK remains to be elucidated. Therefore, the collective effect of MCAK and XMAP215 on microtubule dynamics remains an intriguing question.

To investigate this issue, we studied how MCAK and XMAP215 together regulate microtubule grow rate and catastrophe frequency. Based on our pilot experiments, we chose 20 nM for MCAK and 50 nM for XMAP215 as representative concentrations. In the presence of 20 nM MCAK, dynamic microtubules could grow, but their catastrophe frequency was significantly increased, and their lifetime was shortened (**Fig. 7 A-B**). In contrast, 50 nM XMAP215 increased the growth rate more than fourfold (control: $0.8 \pm 0.1 \mu\text{m min}^{-1}$, $n=97$ microtubules from 6 assays; 50 nM XMAP215: $3.7 \pm 0.3 \mu\text{m min}^{-1}$, $n=50$ microtubules from 3 assays). Notably, XMAP215 alone also caused a mild shortening of microtubule lifetime (**Fig. 7A-B**) (50 nM XMAP215: 264.6 ± 12.9 seconds, $n=237$ growing events from 3 assays), as compared to that of the control group (323.4 ± 12.6 seconds of control, $n=443$ growing events from 6 assays; $p>1$, the two-tailed Mann-Whitney U test with Bonferroni correction). This is consistent with previous reports suggesting that XMAP215 enhances fluctuations in microtubule end structure (Farmer et al., 2021). When both 20 nM MCAK and 50 nM XMAP215 were added, the microtubule growth rate was comparable to that observed with XMAP215 alone ($3.9 \pm 0.3 \mu\text{m min}^{-1}$, $n=52$ microtubules from 3 assays; $p=0.5$, the two-tailed Mann-Whitney U test with Bonferroni correction), suggesting that MCAK does not significantly affect XMAP215's function as a polymerase. However, the lifetime of dynamic microtubules (170.3 ± 7.5 seconds, $n=283$ growing events from 3 assays) was shorter than that observed with XMAP215 alone ($p=2 \times 10^{-6}$, the two-tailed Mann-Whitney U test with Bonferroni correction), while it was similar to that observed with MCAK alone (196.7 ± 6.6 seconds, $n=621$ growing events from 6 assays; $p>1$, the two-tailed Mann-Whitney U test with Bonferroni correction) (**Fig. 7B**). To determine whether this reflects the simple summation of the individual contributions of MCAK and XMAP215 or a synergistic effect, we calculated the probability-based catastrophe frequency for all conditions (**Fig. 7C**) (Gardner et al., 2011). We found that the effects of MCAK and XMAP215 on the probability of microtubule catastrophe were additive, indicating that there was no synergistic effect on catastrophe frequency (**Fig. 7C**). This suggests that, at least at this concentration combination, MCAK and XMAP215 independently regulate different parameters of microtubule dynamics, namely catastrophe frequency and growth rate, while simultaneously interacting with microtubule ends.

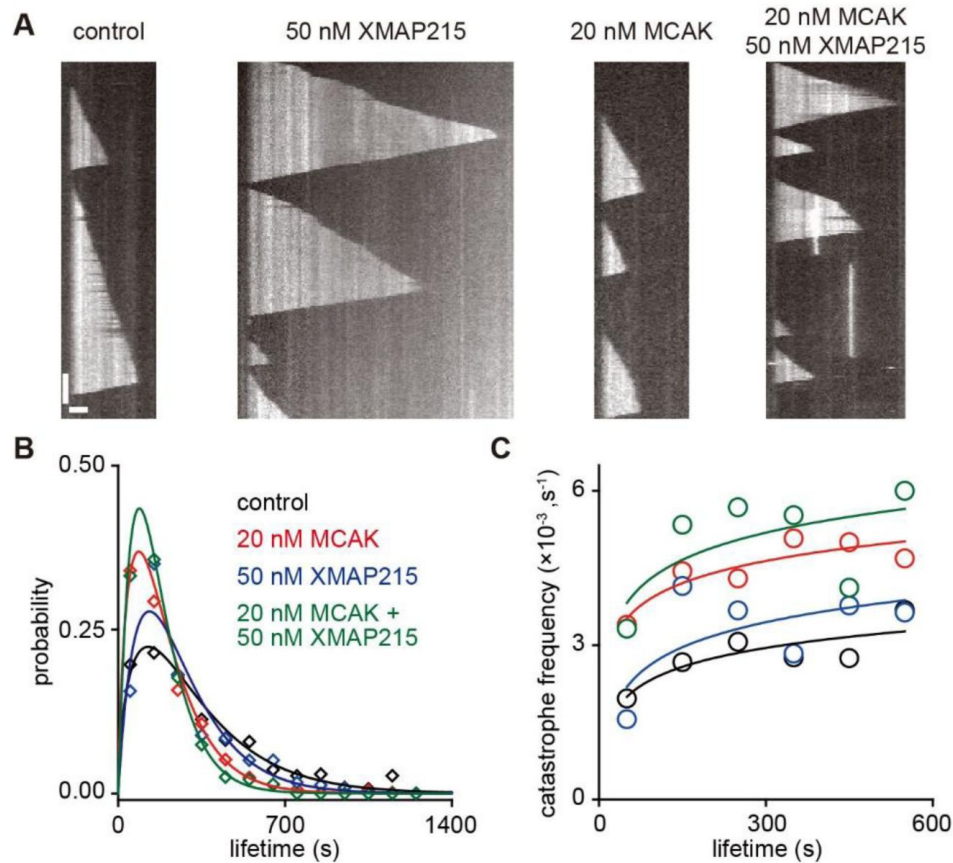


Figure 7.

Functional specification of MCAK and XMAP215 at growing microtubule ends

(A) Representative kymographs of dynamic microtubules (tubulin: 10 μM) under four conditions: control (no MAPs), 20 nM MCAK (1 mM ATP), 50 nM XMAP215 or both (1 mM ATP). Scale bars: vertical, 100 s; horizontal, 2 μm .

(B) Probability distribution of the microtubule lifetime under the conditions indicated in the panel **A**. Lines represent gamma function fits. Control: $n=443$ microtubules from 7 assays. 20 nM MCAK: $n=621$ microtubules from 7 assays. 50 nM XMAP215: $n=237$ microtubules from 3 assays. 20 nM MCAK+50 nM XMAP215: $n=283$ microtubules from 3 assays.

(C) The probability-based catastrophe frequency versus microtubule lifetime showing how the likelihood of catastrophe depended on the age of microtubules. Data derived from panel **A**.

Discussion

This study provides mechanistic insights into the end-binding mechanism of MCAK. The previous established view on the end-binding mechanism of MCAK is based on its specific recognition of curved protofilaments at the distalmost end of growing microtubules (Asenjo et al., 2013 [DOI](#); Benoit et al., 2018 [DOI](#); Tan et al., 2008 [DOI](#)). The findings presented here expand on this model by showing that MCAK also binds to the EB cap, a binding preference that likely facilitates both end-binding and the subsequent depolymerizing activity of MCAK. Below, we discuss our key findings and their implications.

MCAK binds to the EB cap at growing microtubule ends

Our primary discovery is that MCAK preferentially binds to the EB cap, where the GDP-Pi-tubulins accumulate. This conclusion is based on four main lines of evidence. First, localization analysis demonstrates that MCAK binds to the entire GTP cap, including both the EB cap and the distalmost cap. Second, MCAK·ADP and MCAK·APO show the significantly reduced, but not abolished, binding preference for growing microtubule ends, while their binding preference for curved protofilaments is absent. These different binding behaviors suggest that additional structural features, beyond curved protofilaments, contribute to MCAK's growing end-binding preference. Third, MCAK preferentially binds to GTPγS microtubules over GDP microtubules. Since the tubulins in GTPγS microtubules are considered to mimic, at least, some of the key features of the GDP-Pi-tubulins in the EB cap (Maurer et al., 2011 [DOI](#); Maurer et al., 2012 [DOI](#)), this suggests that MCAK can recognize the molecular features of tubulin dimers within the EB cap. Fourth, previous cryo-EM studies have shown that GTP hydrolysis leads to longitudinal compression of tubulin dimers, increasing the contact area at the interdimer interface (LaFrance et al., 2022 [DOI](#); Manka and Moores, 2018 [DOI](#); Zhang et al., 2015 [DOI](#)). Therefore, the interface in GTPγS microtubules is characteristic, being larger than that of a straight GMPCPP-tubulin lattice but smaller than that of a straight GDP-tubulin lattice. We find that the L2 loop contributes to the binding-preference to the EB cap, and it was previously reported that the L2 loop of MCAK directly contacts the interdimer interface (**Fig. 6A** [DOI](#)) (Trofimova et al., 2018 [DOI](#); Wang et al., 2017 [DOI](#)), potentially providing a structural basis for MCAK to distinguish the nucleotide-dependent features of tubulin dimers.

The end-binding affinity of MCAK

In our experiments, we observed that MCAK exhibits a relatively low end-binding affinity (69 μM), raising the question of its physiological relevance. Several factors may contribute to this observation. First, the in vitro system used may not fully recapitulate physiological conditions. For instance, lowering the salt concentration significantly increases binding affinity (Helenius et al., 2006 [DOI](#); Maurer et al., 2011 [DOI](#); McHugh et al., 2019 [DOI](#)). Additionally, while numerous binding events with short durations were detected, we excluded transient interactions from our analysis to facilitate quantification. This likely led to an underestimation of the on-rate and, consequently, the binding affinity. Furthermore, previous studies have shown that retaining a His-tag markedly enhances microtubule binding (Maurer et al., 2011 [DOI](#); Zhu et al., 2009 [DOI](#)). To minimize interference from purification tags, we ensured complete removal of the His-tag during protein preparation (**Fig. S1** [DOI](#)), which may explain the lower apparent binding affinity in our study compared to others. Finally, a low affinity is not necessarily unexpected. Considering the microtubule end as a receptor with multiple binding sites for MCAK, the overall binding affinity is in the nanomolar range (260 nM). Indeed, only a few MCAK molecules may be sufficient to induce microtubule catastrophe.

A natural follow-up question is the physiological relevance of MCAK's direct binding to microtubule ends compared to the well-established EB1-mediated pathway (Gouveia et al., 2010; Honnappa et al., 2009; Lee et al., 2008; McHugh and Welburn, 2023). The relative contributions of these two pathways depend on MCAK's end-binding affinity. To estimate their proportional contributions, we employed a simple model (Supplementary Appendix-2 and Fig. S7). The modeling results suggest that MCAK's direct end-binding affinity contribute through two distinct mechanisms (Supplementary Appendix-2). First, MCAK establishes an EB1-independent regulatory pathway. Simultaneously, it enhances the EB1-dependent pathway by increasing the microtubule-end affinity of the MCAK-EB1 complex. Notably, these two mechanisms are not mutually exclusive. Particularly intriguing is that the direct regulatory pathway may account for previously observed experimental phenomena. For example, prior studies have shown that both MCAK and EB1 accumulate at the centromere and centrosome during cell division (Bahmanyar et al., 2009; Fodde et al., 2001; Kline-Smith et al., 2004; Maney et al., 1998; Wordeman and Mitchison, 1995). However, EB1 neither influences MCAK's localization at these sites nor significantly modulates its local functions (Domnitz et al., 2012).

Implications for the working mechanism of MCAK

We think that our findings add to the current working model of MCAK in several ways. First, the end-binding preference of MCAK for growing microtubule ends is nucleotide state-dependent. The present study demonstrates that MCAK exhibits the strongest end-binding preference in the ATP-bound state, suggesting that MCAK·ATP could more effectively distinguish between the end and the lattice. This supports the view that MCAK acts at the growing microtubule ends in the ATP-bound state (Asenjo et al., 2013; Benoit et al., 2018; Desai et al., 1999; Tan et al., 2008; Wang et al., 2015).

Second, the binding preference of MCAK to the GDP·Pi-tubulins provides a mechanism for MCAK remaining bound to the GTP cap after ATP hydrolysis. In a speculative scenario (Fig. 8A), after the terminal tubulin dimers are removed, the MCAK motor domain transitions to the ADP state and loses its affinity for curved protofilaments. However, it still preferentially binds to GDP·Pi-tubulins, allowing MCAK to stay within the GTP cap. As ADP is replaced by a new ATP, the motor domain could locally diffuse to explore a new working site for the next round of catalysis, which may facilitate MCAK's processivity (Cooper et al., 2010; Friel and Howard, 2011; Helenius et al., 2006; Hunter et al., 2003).

Third, we found that XMAP215 and MCAK independently regulate the growth rate and catastrophe frequency of microtubules, respectively (Fig. 7 and 8B). Previous studies have suggested that XMAP215 and MCAK exert antagonistic effects on microtubule assembly (Barr and Gergely, 2008; Kinoshita et al., 2001; Moriwaki and Goshima, 2016; Tournebize et al., 2000), possibly through competition for binding sites. Structural studies have indicated overlap in the binding sites of MCAK and XMAP215 on tubulin dimers (Ayaz et al., 2012; Trofimova et al., 2018), lending support to this hypothesis. However, in the present study, we observed that the binding regions of XMAP215 and MCAK at growing microtubule ends are not entirely overlapping, with MCAK predominantly binding proximally to the binding region of XMAP215. These findings suggest an alternative mechanism for the combined effects of XMAP215 and MCAK. In this model, MCAK and XMAP215 independently regulate catastrophe frequency and growth rate, respectively, modulating microtubule dynamics through additive effects, rather than through direct competition for binding sites.

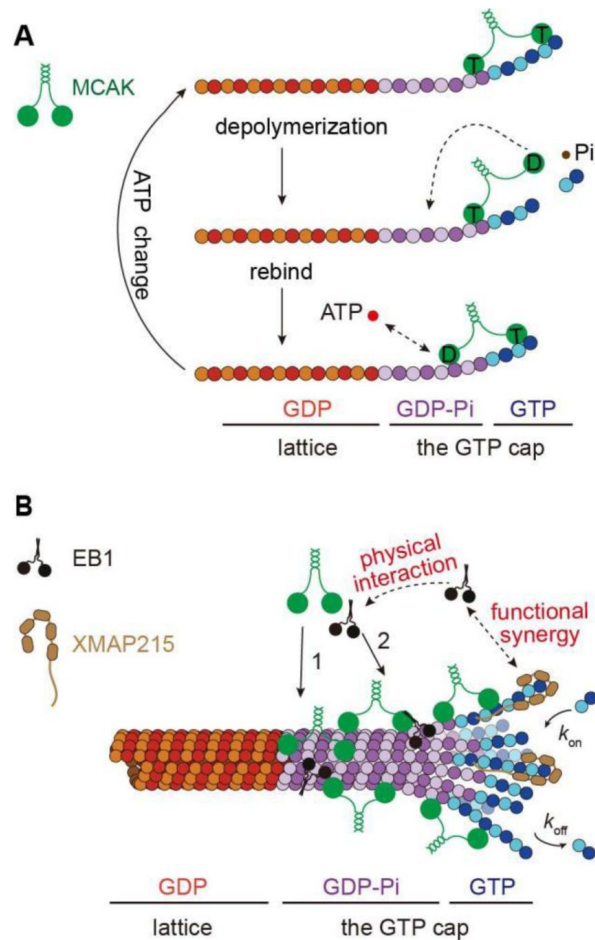


Figure 8.

Cartoon schematics depicting the working model of MCAK at growing microtubule ends

(A) A hypothetical model for the binding cycle of MCAK, given its binding preference on the EB cap. Arrows indicate the transitions in nucleotide-dependent MCAK functional states. T: MCAK·ATP. D: MCAK·ADP.

(B) Cooperation schematics of MCAK, EB1 and XMAP215 at growing microtubule ends. Pathway 1: the direct binding of MCAK to microtubule ends. Pathway 2: the indirect binding of MCAK to microtubule ends via EB1.

Materials and Methods

Protein expression and purification

MCAK purification

GFP-MCAK, its mutants and MCAK-RFP (construct schematics shown in **Fig. s1**) were expressed using a modified BAC-to-BAC protocol. Purification was performed with adaptations based on previously described methods (Helenius et al., 2006). Briefly, the Sf9 cells were lysed by Dounce homogenization with ice-cold lysis buffer. The lysate was centrifuged at 40,000 rpm for 60 min at 4°C (XPN-100 ultracentrifuge, Type 45 Ti rotor, Beckman, USA). Cleared supernatant was filtered through a 0.45 µm filter (Pall, 66229) and then purified using a cation-exchange column (HiTrap SP-HP, Cat.17115101, GE, USA). The protein was eluted using the elution buffer supplemented with a continuous salt gradient (0.15~1.0 M NaCl). Peak fractions were pooled together and then subjected to a Ni-sepharose column (QIAGEN, 30210). The column was then washed with the washing buffer I. After the washing step, 3C Protease (final concentration 10 µg/ml) (Thermo Fisher Scientific, 88946) was employed to perform on-column his-tag cleavage overnight at 4°C. The successful removal of His-tag was subsequently confirmed using Western blot analysis (**Fig. s1**). Cleaved protein was further purified through size-exclusion chromatography (Superdex 200 increase 10/300 GL column, Cat. 28990944, Cytiva, USA) and eluted with the washing buffer II. Purified MCAK aliquots were frozen in liquid N₂ and stored at -80°C.

Lysis buffer: 50 mM HEPES pH 7.5, 150 mM NaCl, 5% glycerol, 0.1% Tween 20, 1.5 mM MgCl₂, 3 mM EGTA, 1 mM DTT, 0.5 mM Mg-ATP, 10 units/ml Benzonase. Elution buffer: 6.7 mM HEPES pH 7.5, 6.7 mM MES, 6.7 mM sodium acetate, 1.5 mM MgCl₂, 10 µM Mg-ATP. Washing buffer I: 50 mM NaPO₄ buffer pH 7.5, 300 mM NaCl, 10 mM imidazole, 10% glycerol, 1 mM MgCl₂, 10 µM Mg-ATP. Washing buffer II: BRB80, 300 mM KCl, 1 mM DTT and 10 µM Mg-ATP. BRB80: 80 mM PIPES/KOH pH 6.9, 1 mM MgCl₂, 1 mM EGTA.

EB1 purification

EB1-GFP (construct schematic shown in **Fig. s1**) was expressed and purified as previously described (Song et al., 2020). Briefly, EB1-GFP was expressed in the *BL21 E. coli* strain and lysed by sonication with ice-cold lysis buffer. The lysate was centrifuged at 40,000 rpm for 60 min at 4°C. Cleared supernatant was filtered through a 0.45 µm filter and loaded onto a Ni-sepharose column. The column was then washed using the washing buffer. After the washing step, 3C Protease (final concentration 10 µg/ml) was employed to perform on-column his-tag cleavage overnight at 4°C. Eluates were desalted into storage buffer using the PD-10 desalting column (GE, 17085101). The final purified protein aliquots were frozen in liquid N₂ and stored at -80 °C.

Lysis buffer: 50 mM NaPO₄ buffer pH 7.5, 300 mM NaCl, 10% glycerol, 10 mM imidazole, 1 mM DTT, 0.1% Tween 20. Washing buffer: 50 mM NaPO₄ buffer pH 7.5, 300 mM NaCl, 10% glycerol, 30 mM imidazole, 1 mM DTT. Storage buffer: BRB80, 100 mM KCl, 10% glycerol and 1 mM DTT.

XMAP215 purification

XMAP215-GFP-his₆ (construct schematic shown in **Fig. s1**) was expressed using a modified BAC-to-BAC protocol. The purification was performed as previously described (Brouhard et al., 2008). Briefly, the cells were lysed by Dounce homogenization with ice-cold lysis buffer. The lysate was centrifuged at 40,000 rpm for 60 min at 4°C. Supernatant was filtered through a 0.45 µm filter and then purified using a cation-exchange column. The protein was eluted using the elution buffer supplemented with a continuous salt gradient (0.15~1.0 M NaCl). Peak fractions were pooled

together and then loaded onto a Ni-sepharose column. The column was then washed with the washing buffer I. The protein was eluted using the washing buffer I supplemented with a continuous imidazole gradient (20–300 mM). Eluted fractions with best purity were pooled together and further purified through size-exclusion chromatography (Superdex 200 increase 10/300 GL column, Cat. 28990944, Cytiva, USA) and eluted using the washing buffer II. The final purified XMAP215 was frozen in liquid N₂ and stored at –80°C.

Lysis buffer: 50 mM HEPES pH 7.5, 50 mM NaCl, 5% glycerol, 0.1% Tween 20, 1 mM DTT, 10 units/ml Benzonase. Elution buffer: 6.7 mM HEPES pH 7.5, 6.7 mM MES, 6.7 mM sodium acetate. Washing buffer I: 50 mM NaPO₄ buffer pH 7.5, 300 mM NaCl, 10 mM imidazole, 10% glycerol, 1 mM MgCl₂, 10 μM Mg-ATP. Washing buffer II: BRB80, 150 mM KCl, 1 mM DTT.

Tubulin preparation

Tubulin was purified from porcine brain tissues through two cycles of temperature-dependent polymerization and depolymerization followed by affinity purification using the TOG-based affinity column, as previously described (Gell et al., 2010 [↗](#); Widlund et al., 2012 [↗](#)). Purified tubulin was labeled with biotin (Thermo Fisher Scientific, 20217), TAMRA (Thermo Fisher Scientific, C1171) and Alexa Fluor 647 (Thermo Fisher Scientific, A2106) using the NHS esters according to the standard protocols (Gell et al., 2010 [↗](#)).

Size-exclusion chromatography

To confirm the oligomeric state of the purified proteins, size-exclusion chromatography analysis (Superose™ 6 Increase 5/150 GL column, Cat. 29091597, Cytiva, USA) was performed using BRB80 supplemented with 50 mM KCl and 1 mM ATP.

Microtubule depolymerization assay

Microtubules (5% Alexa Fluor 647 labeled and 20% biotin labeled) were polymerized in the presence of GMPCPP (JenaBioscience, NU-405L) as previously described (Song et al., 2020 [↗](#)). The GMPCPP-stabilized microtubules were immobilized on the surface of a cover glass using the biotin-NeutrAvidin protein links (Thermo Fisher Scientific, 31000). GFP-MCAK was then added into the flow cell in the imaging buffer. The sample was kept at 35°C using a temperature controller (Tokai Hit, Japan).

Images were recorded using a total internal reflection fluorescence microscope (TIRFM) (Olympus, Japan) equipped with a 100×1.49 N.A. oil TIRF objective (Olympus, Japan) and an Andor 897 Ultra EMCCD camera (Andor, Belfast, UK). Images were recorded every 5 s with a 100 ms exposure. Imaging buffer: BRB80 supplemented with 1 mM ATP, 50 mM KCl, 80 mM D-glucose, 0.4 mg/ml glucose oxidase, 0.2 mg/ml catalase, 0.8 mg/ml casein, 1% β-mercaptoethanol, 0.001% Tween 20.

Microtubule dynamics assay

Microtubule dynamics assay was performed as previously described (Song et al., 2020 [↗](#)). Briefly, the immobilized GMPCPP-stabilized microtubules were used as the template for microtubule growth. Tubulin dimers (13% TAMRA labeled) and the protein of interest were then added into the flow cell in the imaging buffer. The sample was kept at 35°C using a temperature controller (Tokai Hit, Japan). Images were recorded using a TIRFM (Olympus, Japan). To record microtubule dynamics, images were recorded every 5 s with a 100 ms exposure. To record single molecule binding events, images were recorded every 100 ms with a 50 ms exposure. Imaging buffer: BRB80 supplemented with 1 mM ATP, 2 mM GTP, 50 mM KCl, 0.15% sodium carboxymethylcellulose, 80 mM D-glucose, 0.4 mg/ml glucose oxidase, 0.2 mg/ml catalase, 0.8 mg/ml casein, 1% β-mercaptoethanol, 0.001% Tween 20.

Screen for single molecules

The average fluorescence intensity over the dwell time of each binding event was used to represent the fluorescence signal of this binding event. Subsequently, a probability distribution of the fluorescence intensities of all binding events was constructed and fitted by Gaussian fitting. A binding event was deemed to represent a single molecule if its intensity fell within the range ($\mu \pm 2\sigma$) of the probability distribution. In addition, only binding events with a dwell time exceeding the duration corresponding to 2 pixels (~ 200 ms) were included in the statistical analysis. As a control for the dimeric GFP-MCAK, we used the fluorescence intensity probability distribution of GFP-MCAK^{SN+M}, a monomeric variant of MCAK (Wang et al., 2012) (Fig. s1).

The diffusion coefficient of MCAK at the lattice of dynamic microtubules

The diffusion coefficient of MCAK on dynamic microtubule lattices was calculated as previously described (Helenius et al., 2006). First, we tracked MCAK's position along its trajectory on dynamic microtubule lattices using the TrackMate plugin in ImageJ (Fiji) (Tinevez et al., 2017), maintaining a minimum track length of 400 ms (100 ms/frame). Next, the mean-squared displacement (MSD) was computed from 203 trajectories across three assays using MATLAB. Finally, MSD values were plotted against the time interval, and the diffusion coefficient (D) was determined by linear regression analysis ($\langle x^2 \rangle = 2Dt$) using Origin 8.0 (OriginLab Corporation, USA).

The flux of MCAK reaching the growing microtubule ends

The flux of MCAK reaching growing microtubule ends was calculated by a one-dimensional lattice-diffusion model, based on Fick's first equation of diffusion (Helenius et al., 2006):

$$J = -D \cdot \partial c / \partial x \quad \text{Eq. 1}$$

where J represents the flux of MCAK reaching growing microtubule ends, $c(x, t)$ is the concentration of MCAK on microtubule lattice at position x from the growing microtubule end and at time t .

$$\text{When } x = 0 \quad J_0 = -D \cdot c_{\infty} / x_0 \quad \text{Eq. 2}$$

where

$$x_0 = (D / k_{\text{off}})^{1/2} \quad \text{Eq. 3}$$

$$c(x \rightarrow \infty) = c_{\infty} = k_{\text{on}} \cdot C / k_{\text{off}} \quad \text{Eq. 4}$$

where C is the concentration of MCAK. Therefore, the flux was calculated by taking the measured parameters (D , k_{on} , C and k_{off}) into the equations.

Localization analysis at growing microtubule ends

The single-molecule localization at growing microtubule ends was conducted as previously described (Maurer et al., 2014; Song et al., 2020). Briefly, Fluorescence calibration of the captured movie was performed using fluorescent beads (T7279, Thermo Fisher Scientific). During the dwell time of a single-molecule binding event, the microtubule lattice was considered as a Gaussian wall and its end as a half-Gaussian in every frame. The peak position of the half-Gaussian was used as a reference to measure the intensity profile of single-molecule binding events frame by frame. A Gaussian function was employed to fit the intensity profile of individual

molecules. The position at the peak of the intensity profile was taken as the single molecule's location in every frame (**Fig. s2**). Subsequently, the average of these positions was taken to represent the single-molecule binding event's position during its dwell time (**Fig. s2**). This procedure was iterated for all single-molecule binding events to determine their positions during their respective dwell times. Then, the spatial positional distributions of GFP-MCAK and XMAP215-GFP were fitted by Gaussian fitting and the spatial positional distribution of EB1-GFP was fitted to an exponentially modified Gaussian function, from which we calculated the FWHM as an estimation for their length of the binding regions (**Fig. 2C**).

Location precision analysis

The observed image results from the convolution of the fluorescent object with the system's point spread function (PSF). Consequently, the standard deviation of the observed image reflects a combination of measurement error (standard deviation) in determining the object's position and the width of the PSF (**Eq. 5**) (Bohner et al., 2016; Demchouk et al., 2011). To characterize the PSF of the imaging system, we used multicolor fluorescent beads (T7279, ThermoFisher Scientific) with a 100 nm diameter (Hirvonen et al., 2009). When measuring the position of a microtubule end in a single image, we modeled the microtubule lattice as a Gaussian wall and its end as a half-Gaussian. Using this fitting approach, we estimated the measurement error (standard deviation) of the convolved microtubule end positions and deconvolved the system's PSF contribution using **Eq. 5**. The mean standard deviation of end positions over the dwell time of a binding event was then used as the measurement error for the microtubule end position. This procedure was iterated for all microtubule ends with single-molecule binding events to construct their measurement errors. Then, the probability distribution of measurement errors was fitted by Gaussian fitting. The standard deviation of the probability distribution was taken as the precision in estimating microtubule end positions (~1.3 nm). The same calculation method was applied to determine the positional precision of other single molecules (~18 nm).

$$\sigma_I^2 = \sigma_0^2 + \sigma_{\text{PSF}}^2 \quad \text{Eq. 5}$$

Microtubule polymerization

GMPCPP, GDP and GTPγS microtubules were polymerized as previously described (Manka and Moores, 2018). Briefly, GMPCPP-microtubules were polymerized using a mixture of 10 μM tubulin, 1 mM GMPCPP (JenaBioscience, NU-405L) and 4 mM MgCl₂ in BRB80, which was incubated for 2 hours at 37°C. The polymerized microtubules were then pelleted using an Air-Driven Ultracentrifuge (Beckman, 340401), resuspended in BRB80 and stored at 37°C. GDP-microtubules were polymerized using a mixture of 40 μM tubulin, 1 mM GTP (Roche, 10106399001), 4 mM MgCl₂ and 4% DMSO (Sigma, 276855). The mixture was incubated for 30 min at 37°C. The polymerized microtubules were collected using an Air-Driven Ultracentrifuge, resuspended in BRB80 with 20 μM taxol (Cell Signaling Technology, 9807) and stored at 37°C. GTPγS microtubules were prepared using two rounds of polymerizations. In the first round, a mixture of 40 μM tubulin, 2 mM GTPγS (Roche, 10220647001) and 4 mM MgCl₂ in BRB80 was incubated for 5 hours at 37°C. In the second round, 5 μl of the product from the first polymerization was mixed with additional 40 μM tubulin, 2 mM GTPγS and 4 mM MgCl₂ to create a final 25 μl reaction mixture. This mixture was incubated overnight at 37°C to obtain long GTPγS microtubules. The polymerized microtubules were collected using an Air-Driven Ultracentrifuge, resuspended in BRB80 with 20 μM taxol and stored at 37°C.

Microtubule-Stimulated ATPase Assay

The microtubule stimulated ATPase activities of MCAK and its mutants were performed by using the Kinesin ATPase End-Point Biochem Kit (Cytoskeleton, Inc., BK053). The measurements were based on the malachite green phosphate assay to probe inorganic phosphate generated during the

reaction. Wild-type MCAK served as the control, and each type of variant was subjected to multiple replicate measurements.

Binding of MCAK to different microtubules

The microtubules in different nucleotide states were first immobilized in flow cells. GFP-MCAK or its mutants were then added into the flow cell in the imaging buffer. Images were recorded every 300 ms with a 100 ms exposure. Imaging buffer: BRB80 supplemented with 1 mM ATP, ADP or AMPPNP, 20 μ M taxol, 50 mM KCl, 80 mM D-glucose, 0.4 mg/ml glucose oxidase, 0.2 mg/ml catalase, 0.8 mg/ml casein, 1% β -mercaptoethanol, 0.001% Tween 20.

The binding intensity of GFP-MCAK at GMPCPP microtubule ends

To measure the binding intensity of GFP-MCAK at GMPCPP microtubule ends, an intensity profile of GFP-MCAK or the GMPCPP microtubule was plotted along the microtubule. The average binding intensity of GFP-MCAK of three pixels represents the end-binding intensity of GFP-MCAK at the corresponding position of the end, to the lattice from the midpoint of the microtubule intensity attenuation at the microtubule end.

Gamma fitting of microtubule lifetimes and catastrophe frequency analysis

The lifetime data of dynamic microtubules were fitted to the gamma density function using the dfittool toolbox in MATLAB (Mathworks, USA). The probability-based catastrophe frequency was calculated as the ratio of the number of catastrophe events observed at a certain lifetime to the total number of microtubules that reached this lifetime (Gardner et al., 2011 [link](#)).

Supplementary appendix-1

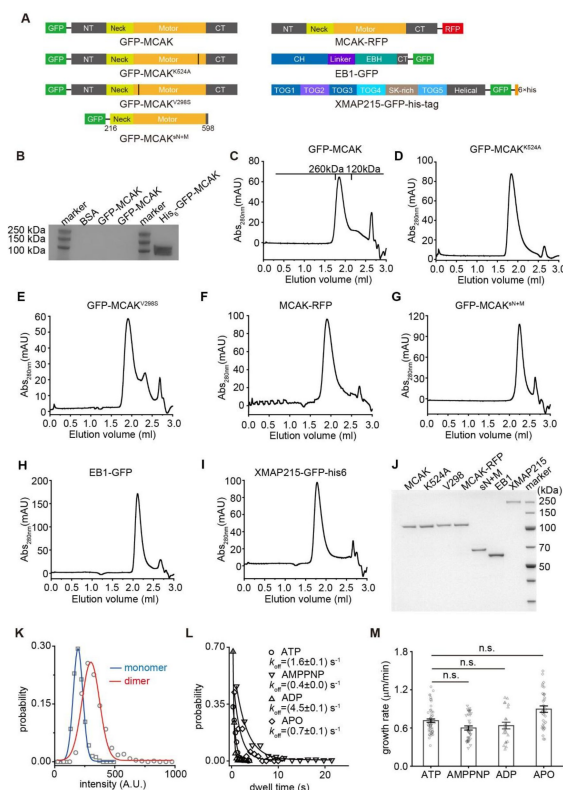


Figure S1.

Biochemical and functional characterization of purified MCAK, its variants, EB1 and XMAP215

(A) Domain architecture schematics of GFP-MCAK and its mutants, MCAK-RFP, EB1-GFP, and XMAP215-GFP used in this study.

(B) The western blot result showing the removal of His-tag in purified GFP-MCAK. In the figure, the molecular weight marker lane and the protein sample lane were merged. The molecular weight of GFP-MCAK monomer is 110 kDa. Anti-his antibody was used for detection.

(C-I) Size-exclusion chromatography (Superose™ 6 Increase 5/150 GL) profiles of GFP-MCAK, GFP-MCAK^{K524A}, GFP-MCAK^{V298S}, MCAK-RFP, GFP-MCAK^{SN+M}, EB1-GFP and XMAP215-GFP-his₆. The molecular weight of GFP-MCAK, GFP-MCAK^{K524A}, GFP-MCAK^{V298S} and MCAK-RFP monomer were 110 kDa. The molecular weight of GFP-MCAK^{SN+M} was 73 kDa. The molecular weight of EB1-GFP monomer was 60 kDa. The molecular weight of XMAP215-GFP-his₆ was 260 kDa. The elution volumes of the peak for GFP-MCAK, GFP-MCAK^{K524A}, GFP-MCAK^{V298S}, MCAK-RFP, GFP-MCAK^{SN+M}, EB1-GFP and XMAP215-GFP-his₆ were 1.85 ml, 1.84 ml, 1.91 ml, 1.90 ml, 2.25 ml, 2.11 ml and 1.78 ml, respectively. In panel C, the elution volume of the peak corresponded to the molecular weight range between 120 kDa (EB1-GFP dimer) and 260 kDa (XMAP215-GFP-his₆), suggesting that GFP-MCAK exists as a dimer under this experimental condition.

(J) SDS-PAGE gels of elution fractions corresponding to peaks from size-exclusion chromatography profiles of GFP-MCAK, GFP-MCAK^{K524A}, GFP-MCAK^{V298S}, MCAK-RFP, GFP-MCAK^{SN+M}, EB1-GFP and XMAP215-GFP-his₆ in panels C-I.

(K) Fluorescence intensity distributions of single GFP-MCAK^{SN+M} (blue, n=327 binding events from 3 assays) and GFP-MCAK (red, n=509 binding events from 3 assays) molecules were fitted by Gaussian fitting. The background intensity was subtracted. The concentrations of the two proteins were both 10 nM. Dynamic microtubules (tubulin: 16 μM) were polymerized in the presence of 1 mM ATP. The intensity range-1 (blue, $\mu \pm 2\sigma = 196 \pm 79$ A.U.) and range-2 (red, $\mu \pm 2\sigma = 300 \pm 145$ A.U.) were used to determine if a fluorescence spot represents a single monomer and dimer, respectively.

(L) The apparent off-rate (k_{off}) of GFP-MCAK at growing microtubule ends in the presence of ATP (circle), AMPPNP (down triangle), ADP (up triangle) and APO (diamond). Data of dwell time were from Fig. 1F. k_{off} was calculated by fitting the dwell time of individual GFP-MCAK binding events to a single exponential function.

(M) Statistical quantification of the growth rates of dynamic microtubules with GFP-MCAK in the presence of 1 mM ATP (57 microtubules from 3 assays), AMPPNP (38 microtubules from 2 assays), ADP (24 microtubules from 2 assays) and APO (34 microtubules from 3 assays). Data were from the experiments shown in panel Fig. 1A, which were presented as mean $\pm$ SEM. Statistical analysis was performed using two-tailed Mann-Whitney U test with Bonferroni correction, n.s., no significance, ***, $p < 0.001$.

Figure s2.

Single-molecule localization analysis at growing microtubule ends

(A) Representative kymograph showing a binding event of GFP-MCAK (1 nM, with 1 mM ATP, left) at the growing microtubule end (red, tubulin: 12 μ M). Schematic illustration of MCAK binding to the microtubule end (right). The red cross represented the reference location point. The green dots represented the position of MCAK at every frame during its dwell time. The orange cross indicated the average position of MCAK during the dwell time, which represented the location of MCAK over this period. Scale bars: vertical, 2 s; horizontal, 2 μ m.

(B) Snapshot of a single GFP-MCAK molecule (green), and the microtubule (red) at the timepoint marked by a black arrow in panel A (left). Intensity profiles of the microtubule (red) and the single molecule (green) at this moment. The peak position of the single molecule was determined using Gaussian fitting (green dashed line), which represented the location of the molecule at this moment. Scale bar: 2 μ m.

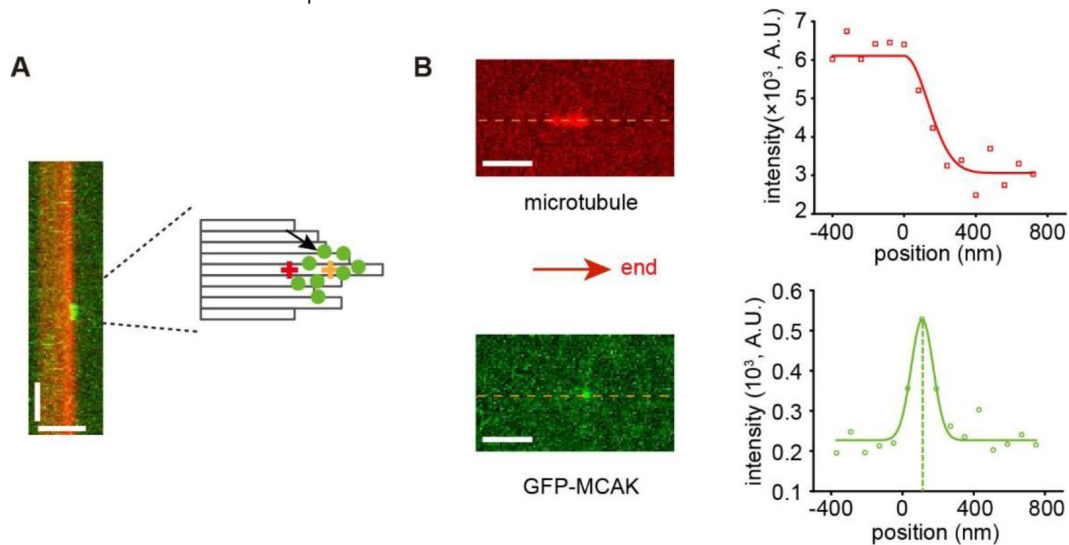


Figure s3.

Projection of single-molecule fluorescence images

Representative images showing how to generate a summation image using 1000 frames of raw images. The binding intensity of GFP-MCAK at GDP versus GTPyS microtubule was indicated using green and yellow arrowheads, respectively. The concentration of GFP-MCAK here was 1 nM. The experiment was performed in the presence of 1 mM AMPPNP. Scale bar: 5 μ m.

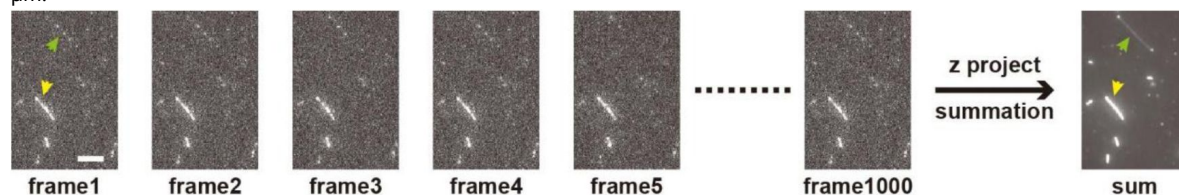


Figure s4.

EB1-GFP strongly binds to GTPyS microtubules

(A) Representative projection images of EB1-GFP binding to GTPyS microtubules. The binding at GDP microtubules was compared to that at GTPyS microtubules in the same flow cell. In GFP channel, the fluorescence intensity of EB1-GFP binding to GTPyS and GDP microtubules were indicated by red and cyan arrowhead, respectively. Scale bar: 5 μ m.

(B) The binding intensity ratios of EB1-GFP on GTPyS microtubules to that on GDP microtubules under various nucleotide conditions. 36 GTPyS/GDP binding ratios from 3 assays. Dashed line on the plot represented 1. All the data were presented as mean $\pm$ SEM. Statistical comparison was performed between the ratios and 1. The comparison was performed using two-tailed Mann-Whitney U test with Bonferroni correction, ***, $p < 0.001$.

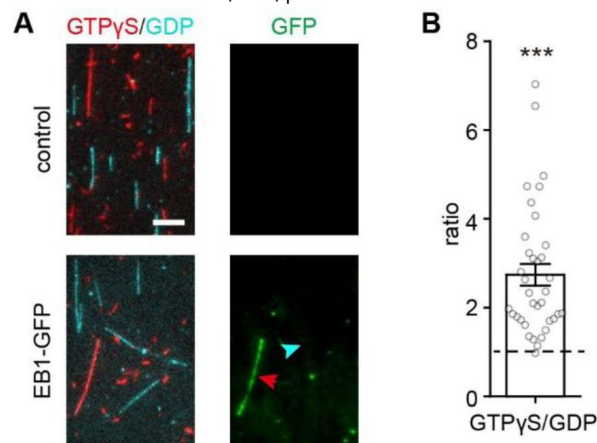


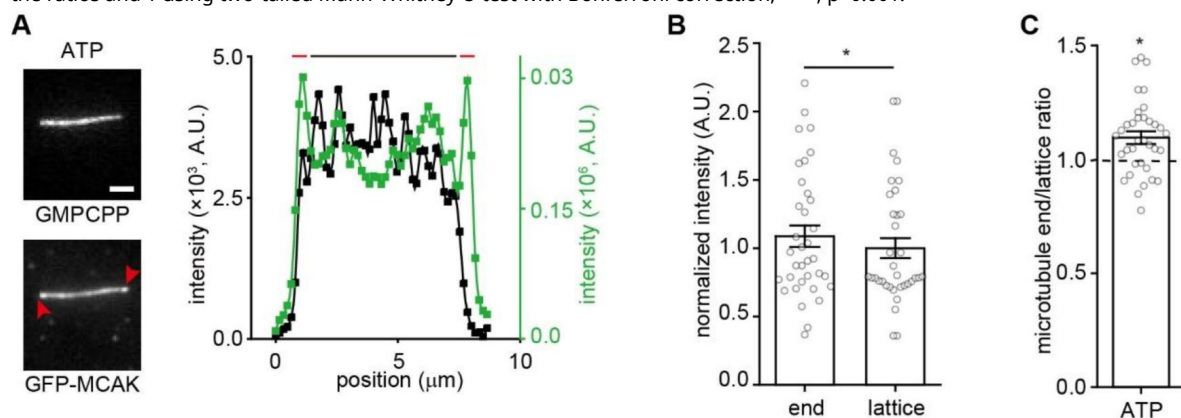
Figure s5.

MCAK preferentially binds to the ends of taxol stabilized GDP microtubules

(A) Representative projection images and intensity profiles showing GFP-MCAK (10 mM) binding at the ends of taxol stabilized GDP microtubules in the presence of 1 mM ATP. Red arrowhead: the end-binding of MCAK events. Intensity profiles (right panels) showed spatial intensity signal distribution of microtubule (black curve) and GFP-MCAK (green curve) along microtubule longitudinal axes from left to right. Red and black bars above the intensity profiles indicated end and lattice regions, respectively. Scale bar=2 μ m.

(B) Statistical quantification of the binding intensity of GFP-MCAK at the end and lattice of taxol stabilized GDP microtubules in the presence of 1 mM ATP (35 microtubules from 2 assays). Data were scaled relative to the binding intensity of GFP-MCAK at the lattice of taxol stabilized GDP microtubules. Data were presented as mean $\pm$ SEM. Statistical analysis was performed using two-tailed pair *t*-test by Bonferroni correction, ***, $p < 0.001$.

(C) The ration of the binding intensity of GFP-MCAK at the end to that on the lattice of GDP microtubules in ATP state. The data were presented as mean $\pm$ SEM. Dashed line on the plot represents 1. Statistical comparisons were performed between the ratios and 1 using two-tailed Mann-Whitney U test with Bonferroni correction, ***, $p < 0.001$.



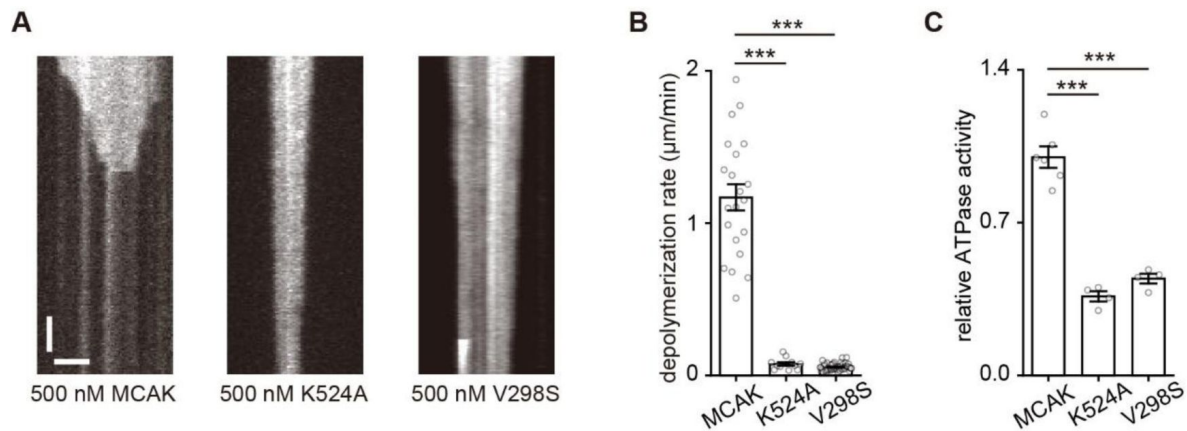


Figure s6.

GFP-MCAK^{K525A} and GFP-MCAK^{V298S} are depolymerizing-deficient and ATPase activity-reduced mutants

- (A) Representative kymographs showing depolymerization of GMPCPP-stabilized microtubules by GFP-MCAK, GFP-MCAK^{K525A} and GFP-MCAK^{V298S} in the presence of 1 mM ATP. Scale bars: vertical, 100 s; horizontal, 2 μm.
- (B) Statistical quantification of microtubule depolymerization rates of GFP-MCAK ($n=21$ microtubules from 3 assays), GFP-MCAK^{K525A} ($n=11$ microtubules from 2 assays) and GFP-MCAK^{V298S} ($n=52$ microtubules from 3 assays). Data were presented as mean $\pm$ SEM. Statistical analysis was performed using two-tailed Mann-Whitney U test with Bonferroni correction, ***, $p<0.001$.
- (C) Microtubule-stimulated ATPase activities of wild-type GFP-MCAK (6 assays), GFP-MCAK^{K525A} (4 assays) and GFP-MCAK^{V298S} (4 assays). Data were normalized by the microtubule-stimulated ATPase activity of the wild-type. Statistical analysis was performed using two-tailed Student's t test, ***, $p<0.001$.

Supplementary appendix-2

Here, we presented a simple model to calculate the relative contribution of the direct and EB-dependent end-binding of MCAK (**Fig. s7** [↗](#)).

Based on the model, we had the dissociation constants:

$$K_0 = \frac{[MCAK] \cdot [EB1]}{[MCAK \cdot EB1]} \quad \text{Eq. 6}$$

$$K_1 = \frac{[MCAK] \cdot [MTE]}{[MCAK \cdot MTE]} \quad \text{Eq. 7}$$

$$K_2 = \frac{[EB1] \cdot [MTE]}{[EB1 \cdot MTE]} \quad \text{Eq. 8}$$

$$K_3 = \frac{[MCAK \cdot EB1] \cdot [MTE]}{[MCAK \cdot EB1 \cdot MTE]} \quad \text{Eq. 9}$$

Then, the relative contribution of the direct and EB-dependent end-binding of MCAK can be expressed as α :

$$\alpha = \frac{[MCAK \cdot MTE]}{[MCAK \cdot EB1 \cdot MTE]} = \frac{K_3 \cdot [MCAK]}{K_1 \cdot [MCAK \cdot EB1]} = \frac{K_3 \cdot K_0}{K_1 \cdot [EB1]} \quad \text{Eq. 10}$$

Here, we considered two scenarios. In the cytoplasm, both EB1 and MCAK undergo free diffusion and can associate with each other without restrictions. The relative concentrations of MCAK and EB1 are critical parameters, but they may vary across different cell types and remain unknown. We also considered the second scenario in which MCAK is locally enriched at specific cellular localizations through an EB-independent mechanism. For example, EB1 does not affect the localization of MCAK at centromere and centrosome, nor does EB1 significantly affect the function of MCAK there (Domnitz et al., 2012 [↗](#)). Here, we assumed that the local concentration of anchored-state MCAK is relatively high; and EB1 remains diffusive and its concentration is nearly constant, as it is continuously replenished in the local space from the vast cytoplasmic pool. In both cases, the ratio of K_3 to K_1 emerges as a key determinant.

K_3 is the end-binding affinity of MCAK-EB1 complex. Intuitively, it depends on the respective microtubule-binding affinities of MCAK and EB1, as well as the cooperativity, if any, of their microtubule-binding behaviors. K_3 can be expressed as:

$$\frac{1}{K_3} = a \frac{1}{K_1} + b \frac{1}{K_2} \quad \text{Eq. 11}$$

$$K_3 = \frac{K_2 \cdot K_1}{aK_2 + bK_1} \quad \text{Eq. 12}$$

where a and b represent the weighting factors of binding sites or cooperativity factors of the binding behaviors. Therefore, K_3 shows a positively correlated, monotonically increasing dependence on K_1 , indicating that the increase in the end-binding affinity of MCAK contributes to that of the MCAK-EB1 complex. Therefore, we think that MCAK's functional impact at microtubule ends derives not only from its intrinsic end-binding capacity, but also its ability to strengthen the EB1-mediated end association pathway.

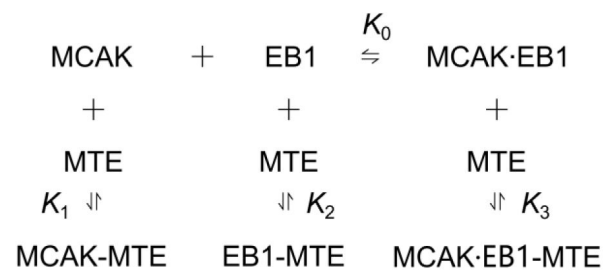


Figure s7.

A simple model for the end-binding of MCAK and EB1.

MCAK can bind to microtubule growing ends through both the direct (left) and EB-dependent (right) pathways. The dissociation constants were K_0 , K_1 , K_2 and K_3 , respectively. MTE: growing microtubule end.

In the simplest case, the formation of the MCAK-EB1-MTE complex arises from the binding of either MCAK or EB1 to microtubule ends; and the binding behaviors for MCAK and EB1 are independent ($a=1$; $b=1$). Consequently, K_3 can be expressed as:

$$K_3 = \frac{K_2 \cdot K_1}{K_2 + K_1} \quad \text{Eq. 13}$$

if $K_1 \ll K_2$, then

$$K_3 \approx K_1 \quad \text{Eq. 14}$$

if $K_1 \gg K_2$, then

$$K_3 \approx K_2 \quad \text{Eq. 15}$$

If $K_1 \approx K_2$, then

$$K_3 \approx \frac{K_1}{2} \quad \text{Eq. 16}$$

In our experiments, we measured the dissociation constants of MCAK to growing microtubule ends is $69 \mu\text{M}$ (K_1). We also performed similar experiments with EB1 and found that EB1 showed the dissociation constant of $722 \mu\text{M}$ (K_2) for growing microtubule ends (**Fig. s8**), similar to the value reported in our previous report (Song et al., 2020). Therefore, substituting Eq. 14 into Eq. 10, we obtained

$$\alpha \approx \frac{K_0}{[EB1]} \quad \text{Eq. 17}$$

Here, if we assume that the cytoplasmic concentration of EB1 is twice the value of K_0 , then $\alpha=0.5$, indicating that 50% of MCAK binds to microtubule ends via the direct binding pathway; even if the EB1 concentration reaches ten times the value of K_0 , 10% of MCAK still utilizes the direct binding pathway. Overall, as the EB1 concentration increases relative to K_0 , α decreases, reflecting a decline in the proportion of MCAK that associates with microtubule ends through the direct binding mechanism.

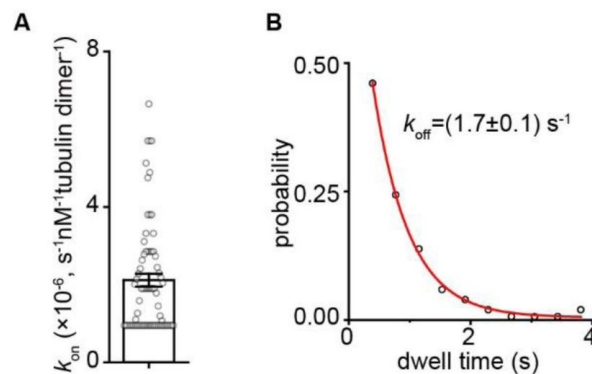


Figure s8.

The binding kinetics of single-molecule EB1-GFP binding to growing microtubule end

(A) Statistical quantification of on-rate (k_{on}) of EB1-GFP's binding to the plus end of dynamic microtubules (data calculated from [Fig. 2](#), $n=73$ microtubules from 3 assays).

(B) The apparent off-rate (k_{off}) of EB1-GFP at growing microtubule ends (data calculated from [Fig. 2](#), $n=153$ bonding events from 3 assays). k_{off} was calculated by fitting the dwell time of individual GFP-MCAK binding events to a single exponential function.

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Author information

Wei Chen

IDG/McGovern Institute for Brain Research, State Key Laboratory of Complex, Severe, and Rare Diseases, School of Life Sciences, Tsinghua University, Beijing, China
ORCID iD: [0000-0001-7454-3882](#)

Yin-Long Song

IDG/McGovern Institute for Brain Research, State Key Laboratory of Complex, Severe, and Rare Diseases, School of Life Sciences, Tsinghua University, Beijing, China
ORCID iD: [0000-0002-1666-9628](#)

Jian-Feng He

IDG/McGovern Institute for Brain Research, State Key Laboratory of Complex, Severe, and Rare Diseases, School of Life Sciences, Tsinghua University, Beijing, China

Xin Liang

IDG/McGovern Institute for Brain Research, State Key Laboratory of Complex, Severe, and Rare Diseases, School of Life Sciences, Tsinghua University, Beijing, China
ORCID iD: [0000-0001-7915-8094](#)

For correspondence: xinliang@tsinghua.edu.cn

Editors

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Reviewer #1 (Public review):

The authors responded to multiple criticisms with additional data and more detailed statistics, in some instances improving the quality of the work. However, I had difficulty understanding some of the authors' responses. The logic was not always apparent, the writing was occasionally confusing or would benefit from more careful wording, and some of the provided responses were superficial or raised new concerns. In some cases, the underlying data needed to support their responses were not shown. Thus, the current version of the manuscript does not sufficiently resolve the following critical issues raised by myself and other reviewers.

(1) A clear new insight into a physiological process or cellular behavior remains lacking. The study largely confirms prior observations of MCAK binding to both the microtubule wall and end. However, it is still unclear whether direct binding to the tip-as opposed to accumulation via wall diffusion or interaction with other tip-binding proteins-is a significant mechanism.

(2) The newly revealed adenosine-nucleotide-dependent binding preferences do not help clarify MCAK's catalytic function or its mechanisms of tip recognition. Consequently, the final summary figure remains speculative and is not convincingly supported by the data. It is also unclear what exactly is meant by the "working model" (figure title), or by the claim of "a simple rule of how the end-binding regulators coordinate their activities" (abstract).

(3) As noted in my previous review, the effects of adding different adenosine nucleotides on MCAK binding to microtubules are much more pronounced than the differences in MCAK binding to tubulin with various guanosine-containing nucleotides, or to lattice versus tip (e.g., Fig. 5E). Therefore, the manuscript title-"MCAK recognizes the nucleotide-dependent feature at growing microtubule ends"-does not do justice to the scale of these effects.

(4) The title implies that MCAK selectively recognizes a feature determined by the tubulin-bound guanosine nucleotide. However, the authors frequently claim that MCAK binds to the "entire GTP cap." It appears that they exclude structural protrusions from their definition of the cap, which is debatable. Even using their definition, the conclusion that MCAK recognizes a specific "nucleotide-dependent feature" seems inconsistent with the claim that it binds uniformly across the cap. These distinctions were not made clear.

(5) Some important technical details are still absent. For example, when reading the authors' response to another reviewer's question, I could not find an explanation of how the k_{on} values for end and wall binding were calculated. These calculations clearly require assumptions, e.g. about the number of binding sites, but these details are not described. In addition, the binding data are expressed in units per tubulin dimer, which are non-standard and make comparisons to other published results difficult. There are other instances where more technical detail would be desirable, but they are too numerous to list here.

(6) Several aspects of data presentation as graphs will make it difficult for other researchers to analyze or interpret the findings. Numerical Excel-style data sheets should be provided for all measurements, including raw data-not just the ratios or derived values shown in plots. Other, more significant issues include use of mean values for non-Gaussian distributions (e.g., dwell times); binding affinities inferred from single-concentration measurements, often under varying conditions (e.g., Figs. 3C, 4); and absence of side-by-side plotted controls (e.g., Fig. 6).

(7) While the authors have added some quantitative values and descriptive detail, the manuscript still lacks a critical comparison of their findings with existing literature. This weakens the impact of the study and limits the reader's ability to place the results in a broader context.

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Reviewer #4 (Public review):

The revised manuscript from Chen et al. implements many of the changes requested by the 3 reviewers of the initial submission. These changes are well-described in the corresponding Response to Reviews document. Of course, not every request from the reviewers was addressed, and the following major concerns remain:

(1) The authors argue that MCAK binds to the same region as EB proteins, which they refer to as the "EB cap". Reviewers asked for experiments that would increase the size of the EB cap to create "comets" (e.g. by increasing the microtubule growth rate); the prediction is that the MCAK signal should increase in size as well. The authors declined to pursue these experiments. As a result, the EB signals and MCAK signals are diffraction-limited spots, as opposed to the predicted exponential decay signals characteristic of EB comets. The various diffraction-limited spots are then aligned with the diffraction-limited signal of the microtubule end. These alignments and sub-pixel comparisons are technically challenging. The revised manuscript does not go far enough to provide compelling evidence that all technical challenges were overcome. Thus, while the authors can safely conclude that MCAK, EBs, and the microtubule end do occupy the same diffraction-limited spot, more precise conclusions are not supported.

(2) The reviewers criticized the initial manuscript for neglecting key references, particularly Kinoshita et al., Science 2001. Indeed, I cannot fathom writing a manuscript about MCAK and XMAP215 without putting a citation to such a landmark paper front and center. The authors have responded by including more discussion of the relevant literature (and citing Kinoshita et al.). However, the revised manuscript is often still cursory in giving credit where credit is due, contextualizing the new data, and generally engaging with the scholarship on MCAK.

(3) The data presented does not include a simple measurement of the impact of MCAK on the catastrophe frequency of microtubules. The authors explain this absence by pointing out that their movies are short (5 min) and high frame rate (10 fps). While I understand that such imaging parameters are necessary to capture single molecule end-binding events, I do not understand why a separate set of experiments could not be performed. This type of "positive control" is often missing, as pointed out by the 3 reviewers.

(4) Salt conditions, protein concentrations, and other key experimental parameters are not varied, even when varying them would provide excellent tests of the authors' hypotheses.

In summary, the revised manuscript is improved in many ways, but the interested reader should look carefully at the previous reviews and compare the measurements presented here with those of other labs.

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Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public Review):

Major concerns:

(1) Is the direct binding of MCAK to the microtubule cap important for its in vivo function?

a. The authors claim that their "study provides mechanistic insights into understanding the end-binding mechanism of MCAK". I respectfully disagree. My concern is that the paper offers limited insights into the physiological significance of direct end-binding for MCAK activity, even in vitro. The authors estimate that in the absence of other proteins in vitro, ~95% of MCAK molecules arrive at the tip by direct binding in the presence of ~ physiological ATP concentration (1 mM). In cells, however, the major end-binding pathway may be mediated by EB, with the direct binding pathway contributing little to none. This is a reasonable concern because the apparent dissociation constant measured

by the authors shows that MCAK binding to microtubules in the presence of ATP is very weak (69 μ M). This concern should be addressed by 1) calculating relative contributions of direct and EB-dependent pathways based on the affinities measured in this and other published papers and estimated intracellular concentrations. Although there are many unknowns about these interactions in cells, a modeling-based analysis may be revealing. 2) the recapitulation of these pathways using purifying proteins in vitro is also feasible. Ideally, some direct evidence should be provided, e.g. based on MCAK function-separating mutants (GDP-Pi tubulin binding vs. catalytic activity at the curled protofilaments) that contribution from the direct binding of MCAK to microtubule cap in EB presence is significant.

We thank the reviewer for the thoughtful comments.

(1) We think that the end-binding affinity of MCAK makes a significant contribution for its cellular functions. To elucidate this concept, we now use a simple model shown in Supplementary Appendix-2 (see pages 49-51, lines 1246-1316). In this model, we simplified MCAK and EB1 binding to microtubule ends by considering only these two proteins while neglecting other factors (e.g. XMAP215). Specifically, we considered two scenarios: one in which both proteins freely diffuse in the cytoplasm and another where MCAK is localized to specific cellular structures, such as the centrosome or centromere. Based on the modeling results, we argue that MCAK's functional impact at microtubule ends derives both from its intrinsic end-binding capacity and its ability to strengthen the EB1-mediated end association pathway.

(2) We agree with the reviewer that MCAK exhibiting a lower end-binding affinity (69 μ M) is indeed intriguing, as one might intuitively expect a stronger affinity, e.g. in the nanomolar range. Several factors may contribute to this observation. First, this could be partly due to the in vitro system employed, which may not perfectly replicate in vivo conditions, especially when considering cellular processes quantitatively. Variations in medium composition can significantly influence the binding state. For example, reducing salt concentration leads to a marked increase in MCAK's binding affinity (Helenius et al., 2006; Maurer et al., 2011; McHugh et al., 2019). Additionally, while numerous binding events with short durations were detected, we excluded transient interactions from our analysis to facilitate quantification. This likely leads to an underestimation of the on-rate and, consequently, the binding affinity. Moreover, to minimize the interference of purification tags (His-tag), we ensured their complete removal during protein sample preparation. Previous studies reported that retaining the His-tag of MAPs affects the binding affinity to microtubules (Maurer et al., 2011; Zhu et al., 2009). Finally, a low affinity is not necessarily unexpected. Considering the microtubule end as a receptor with multiple binding sites for MCAK, the overall binding affinity is in the nanomolar range (260 nM). This does not necessarily contradict MCAK being a microtubule dynamics regulator as only a few MCAK molecules may suffice to induce microtubule catastrophe (as discussed on page 13, lines 408-441).

(3) Ideally, we would search for mutants that specifically interfere with the binding of GDP-Pi-tubulin or the curled protofilaments. However, the mutant we tested significantly impacts the overall affinity of MCAK to microtubules (both end and lattice), making it challenging to isolate and discuss the function of MCAK with respect to the binding to GDP-Pi-tubulin alone. Additionally, we also think that the GDP-Pi-tubulin in the EB cap and the tubulin in the curved protofilaments may share structural similarities. For instance, the tubulin dimers in both states may be less compact compared to those in the lattice, which could explain why MCAK recognizes both simultaneously (Manka and Moores, 2018). However, this remains a conjecture, as there is currently no direct evidence to support it.

b. As mentioned in the Discussion, preferential MCAK binding to tubulins near the MT tip may enhance MCAK targeting of terminal tubulins AFTER the MCAK has been "delivered"

to the distal cap via the EB-dependent mechanism. This is a different targeting mechanism than the direct MCAK-binding. However, the measured binding affinity between MCAK and GMPCPP tubulins is so weak (69 μ M), that this effect is also unlikely to have any impact because the binding events between MCAK and microtubule should be extremely rare. Without hard evidence, the arguments for this enhancement are very speculative.

Please see our response to the comment No. 1. Additionally, we have revised our discussion to discuss the end-binding affinity of MCAK as well as its physiological relevance (please see page 13, lines 408-441; and see Supplementary Appendix-2 in pages 49-51, lines 1246-1316).

(2) The authors do not provide sufficient justification and explanation for their investigation of the effects of different nucleotides in MCAK binding affinity. A clear summary of the nucleotide-dependent function of MCAK (introduction with references to prior affinity measurements and corresponding MCAK affinities), the justifications for this investigation, and what has been learned from using different nucleotides (discussion) should be provided. My take on these results is that by far the strongest effect on microtubule wall and tip binding is achieved by adding any adenosine, whereas differences between different nucleotides are relatively minor. Was this expected? What can be learned from the apparent similarity between ATP and AMPPNP effects in some assays (Fig 1E, 4C, etc) but not others (Fig 1D,F, etc)?

We thank the reviewer for this suggestion. We have revised the manuscript accordingly, and below are the main points of our response

(1) The experiment investigating the effects of different nucleotides on MCAK binding affinity was inspired by the previous studies demonstrating that kinesin-13 interactions with microtubules are highly dependent on their adenosine-bound states. For example, kinesin-13s tightly bind microtubules and prefer to form protofilament curls or rings with tubulin in the AMPPNP state, whereas kinesin-13s are considered to move along the microtubule lattice via one-dimensional diffusion in the ADP-Pi state (Asenjo et al., 2013; Benoit et al., 2018; Friel and Howard, 2011; Helenius et al., 2006). Based on these observations, we wondered whether MCAK's adenosine-bound states might similarly affect its binding preference for growing microtubule ends. We have made the motivation clear in the revised manuscript (please see page 7, lines 199-209).

(2) Our main finding regarding the effects of nucleotides is that MCAK shows differential end-binding affinity and preference based on its nucleotide state. First, MCAK shows the greatest preference for growing microtubule ends in the ATP state, supporting the idea that diffusive MCAK (MCAK-ATP) can directly bind to growing microtubule ends. Second, MCAK-ATP also demonstrates a binding preference for GTPyS microtubules and the ends of GMPCPP microtubules. The similar trends in binding preference suggest that the affinity for GDP-Pi-tubulin and GTP-tubulin likely underpins MCAK's preference for growing microtubule ends. To clarify these points, we have added further discussions in the manuscript (please see page 8, lines 230-233; page9, lines 258-270 and pages 13-14, lines 443-458).

(3) It is not clear why the authors decided to use these specific mutant MCAK proteins to advance their arguments about the importance of direct tip binding. Both mutants are enzymatically inactive. Both show roughly similar tip interactions, with some (minor) differences. Without a clear understanding of what these mutants represent, the provided interpretations of the corresponding results are not convincing.

We thank the reviewer for this comment. In the revised manuscript, we no longer draw conclusions about the importance of end-binding based on the mutant data. Instead, we think that the mutant data provide insights into the structural basis of the end-binding preference.

Therefore, we have rewritten the results in this section to more accurately reflect these findings (please see page 10, lines 295-327).

(4) GMPCPP microtubules are used in the current study to represent normal dynamic microtubule ends, based on some published studies. However, there is no consensus in the field regarding the structure of growing vs. GMPCPP-stabilized microtubule ends, which additionally may be sensitive to specific experimental conditions (buffers, temperature, age of microtubules, etc). To strengthen the authors' argument, Taxol-stabilized microtubules should be used as a control to test if the effects are specific. Additionally, the authors should consider the possibility that stronger MCAK binding to the ends of different types of microtubules may reflect MCAK-dependent depolymerization events on a very small scale (several tubulin rows). These nano-scale changes to tubulins and the microtubule end may lead to the accumulation of small tubulin-MCAK aggregates, as is seen with other MAPs and slowly depolymerizing microtubules. These effects for MCAK may also depend on specific nucleotides, further complicating the interpretation. This possibility should be addressed because it provides a different interpretation than presented in the manuscript.

Regarding the two points raised here, our thoughts are as following

(1) The end of GMPCPP-stabilized microtubules differs from that of growing microtubules, with the most obvious known difference being the absence of the region enriched in GDP-Pi-tubulin. We consider the end of GMPCPP microtubules as an analogue of the distal tip of growing microtubules, based on two key features: (1) curled protofilaments and (2) GMPCPP-tubulin, a close analogue of GTP-tubulin. Notably, both features are present at the ends of both GMPCPP-stabilized and growing microtubules. Moreover, we agree with the suggestion to use taxol-stabilized microtubules as a control. This would eliminate the second feature (absence of GTP-tubulin), allowing us to isolate the effect of the first feature. Therefore, we conducted this experiment, and our data showed that MCAK exhibits only a mild binding preference for the ends of taxol-stabilized microtubules, which is much less pronounced than for the ends of GMPCPP microtubules. This observation supports the idea that GMPCPP-stabilized ends closely resemble the growing ends of microtubules.

(2) The reviewer suggested that stronger MCAK binding to the ends of different types of microtubules might reflect MCAK-dependent depolymerization events on a very small scale. This is an insightful possibility, which we had overlooked in the original manuscript. Fortunately, we performed the experiments at the single-molecule concentrations. Upon reviewing the raw data, we found that under ATP conditions, the binding events of MCAK were not cumulative (see Fig. X1 below) and showed no evidence of local accumulation of MCAK-tubulin aggregates.

Author response image 1.

The representative kymograph showing GFP-MCAK binding at the ends and lattice of GMPCPP microtubules in the presence of 1 mM ATP (10 nM GFP-MCAK), which corresponded to Fig. 5A. The arrow: the end-binding of MCAK. Vertical bar: 1 s; horizontal bar: 2 mm.



(5) It would be helpful if the authors provided microtubule polymerization rates and catastrophe frequencies for assays with dynamic microtubules and MCAK in the presence of different nucleotides. The video recordings of microtubules under these conditions are already available to the authors, so it should not be difficult to provide these quantifications. They may reveal that microtubule ends are different (or not) under the examined conditions. It would also help to increase the overall credibility of this study by providing data that are easy to compare between different labs.

We thank the reviewer for this suggestion. In the revised manuscript, we have provided data on the growth rates, which are similar across the different nucleotide states (Fig. s1). However, due to the short duration of our recordings (usually 5 minutes, but with a high frame rate, 10 fps), we did not observe many catastrophe events, which prevented us from quantifying catastrophe frequency using the current dataset. Since we measured the binding kinetics of MCAK during the growing phase of microtubules, the similar growth rates and microtubule end morphologies suggest that the microtubule ends are comparable across the different conditions.

Reviewer #1 (Recommendations For The Authors):

a. Please provide more details about how the microtubule-bound molecules were selected for analysis (include a description of scripts, selection criteria, and filters, if any). Fig 1A arrows do not provide sufficient information.

We first measured the fluorescence intensity of each binding event. A probability distribution of these intensities was then constructed and fitted with a Gaussian function. A binding event was considered to correspond to a single molecule if its intensity fell within $\mu \pm 2\sigma$ of the distribution. The details of the single-molecule screening process are now provided in the revised manuscript (see page17, lines 574-583).

b. Evidence that MCAK is dimeric in solution should be provided (gel filtration results, controls for Figs1A - bleaching, or comparison with single GFP fluorophore).

In the revised manuscript, we provide the gel filtration results of purified MCAK and other proteins used in this study. The elution volume of the peak for GFP-MCAK corresponded to a molecular weight range between 120 kDa (EB1-GFP dimer) and 260 kDa (XMAP215-GFP-his6), suggesting that GFP-MCAK exists as a dimer (~220 kDa) under experimental condition (please see Fig.s1 and page 5, lines 104-105). In addition, we also measured the fluorescence intensity of both MCAK^{sN+M} and MCAK. MCAK^{sN+M} is a monomeric mutant that contains the neck domain and motor domain (Wang et al., 2012). The average intensity of MCAK^{sN+M} is 196 A.U., about 65% of that of MCAK (300 A.U.). These two measurements suggest that the purified MCAK used in this study exists dimers (see Fig. s1).

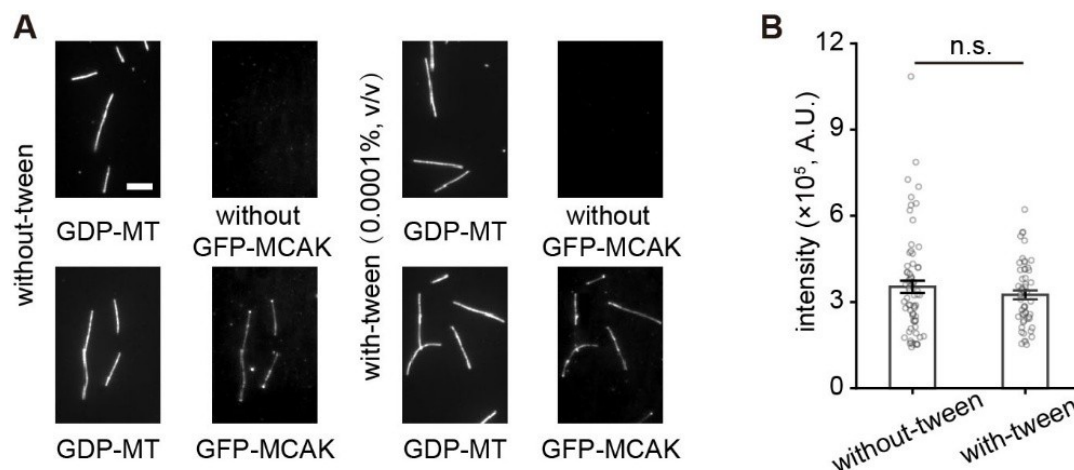
c. Evidence that MCAK on microtubules represents single molecules should be provided (distribution of GFP brightness with controls - GFP imaged under identical conditions). Since assay buffers include detergent, which is not desirable, all controls should be done using the same assay conditions. The authors should rule out that their main results are detergent-sensitive.

(1) Regarding if MCAK on microtubules represent single molecules: please refer to our responses to the two points above.

(2) To rule out the effect of tween-20 (0.0001%, v/v), we performed additional control experiments. The results showed that it has no significant effect on microtubule-binding affinity of MCAK (see Figure below).

Author response image 2.

Tween-20 (0.0001%, v/v) has no significant effect on microtubule-binding affinity of MCAK. (A) The representative projection images of GFP-MCAK (5 nM) binding to taxol-stabled GDP microtubules in the presence of 1 mM AMPPNP with or without tween-20. The upper panel showed the results of the control experiments performed without MCAK. Scale bar: 5 mm. (B) Statistical quantification of the binding intensity of GFP-MCAK binding to GDP microtubules with or without tween-20 (53 microtubules from 3 assays and 70 microtubules from 3 assays, respectively). Data were presented as mean $\pm$ SEM. Statistical comparisons were performed using the two-tailed Mann-Whitney U test with Bonferroni correction, n.s., no significance.



d. How did the authors plot single-molecule intensity distributions? I am confused as to why the intensity distribution for single molecules in Fig 1D and 2A looks so perfectly smooth, non-pixelated, and broader than expected for GFP wavelength. Please provide unprocessed original distributions, pixel size, and more details about how the distributions were processed.

In the revised manuscript, we provided unprocessed original data in Fig. 1B and Fig. 2A. We thank the reviewer for pointing out this problem.

e. Many quantifications are based on a limited number of microtubules and the number of molecules is not provided, starting from Fig 1D and down. Please provide detailed statistics and explain what is plotted (mean with SEM?) on each graph.

We performed a thorough inspection of the manuscript and corrected the identified issues.

f. Plots with averaged data should be supplemented with error bars and N should be provided in the legend. E.g. Fig 1C - average position of MT and peak positions.

We agree with the reviewer. In the revised manuscript, we have made the changes accordingly (e.g. Fig. 2C).

g. Detailed information should be provided about protein constructs used in this work including all tags. The use of truncated proteins or charged/bulky tags can modify protein-microtubule interactions.

We agree with the reviewer. In the revised manuscript, we provide the information of all constructs (see Fig. s1 and the related descriptions in Methods, pages 15-16, lines 476-534).

h. Line 515: We estimated that the accuracy of microtubule end tracking was ~6 nm by measuring the standard error of the distribution of the estimated error in the microtubule end position. - evidence should be provided using the conditions of this study, not the reference to the prior work by others.

i. Line 520: We estimated that the accuracy of the measured position was ~2 nm by measuring the standard error of the fitting peak location". Please provide evidence.

Point h-i: we now provide detailed descriptions of how to estimate tracking and measurement accuracy and error in our work. Please see pages 18-19, lines 626-645.

j. Kymographs in Fig 5G are barely visible. Please provide single-channel greyscale images. What are the dim molecules diffusing on this microtubule?

We have incorporated the changes suggested by the reviewer. We think that some of the dim signals may result from stochastic background noise, while others likely represent transient bindings of MCAK. The exposure time in our experiments was approximately 0.05 seconds; if the binding duration were shorter than this, the signal would be lower (i.e. the “dim” signals). It is important to note that in this study, we selected binding events lasting at least 2 consecutive frames, meaning transient binding events were not included. This point has been clarified in the Methods section (see page 17, lines 573-583).

k. Please provide a methods description for Fig 6. Did the buffer include 1 mM ATP? The presence of ATP would make these conditions more physiological. ATP concentration should be stated clearly in the main text or figure legend.

The buffer contains ATP. In the revised manuscript, we have provided the methods for the experiments of microtubule dynamics assay, as well as the analysis of microtubule lifetimes and catastrophe frequency (see page 17, lines 561-572 and page 20, lines 685-690).

l. Line 104: experiment was performed in BRB80 supplemented with 50 mM KCl and 1 mM ATP, providing a nearly physiological ion strength. Please provide a reference or add your calculations in Methods.

We have provided references on page 5, lines 101-104 of our manuscript.

m. What was the MCAK concentration in Figure 4? Did the microtubule shorten under any of these conditions?

In these experiments, we used a very low concentration of MCAK and taxol-stabilized microtubules, so there's no microtubule shortening observed here. ATP: 10 nM GFP-MCAK; AMPPNP: 1 nM GFP-MCAK; ADP: 10 nM GFP-MCAK; APO state: 0.1 nM GFP-MCAK.

Other criticism:

Text improvements are recommended in the Discussion. For example, line 348: Fourth, the loss of the binding preference.. suggests that the binding preference .. is required for the optimal .. preference.

We thank the reviewer for pointing out this. In the revised manuscript, we conducted a thorough revision and review of the text.

Reviewer #2 (Public Review):

Summary:

In this manuscript, Chen et al. investigate the localization of microtubule kinesin-13 MCAK to the microtubule ends. MCAK is a prominent microtubule depolymerase whose molecular mechanisms of action have been extensively studied by a number of labs over the last ~twenty years. Here, the authors use single-molecule approaches to investigate the precise localization of MCAK on growing microtubules and conclude that MCAK preferentially binds to a GDP-Pi-tubulin portion of the microtubule end. The conclusions are speculative and not well substantiated by the data, making the impact of the study in its current form rather limited. Specifically, greater effort should be made to define the region of MCAK binding on microtubule ends, as well as its structural characteristics.

Given that MCAK has been previously shown to effectively tip-track growing microtubule ends through an established interaction with EB proteins, the physiological relevance of the present study is unclear. Finally, the manuscript does not cite or properly discuss a number of relevant literature references, the results of which should be directly compared and contrasted to those presented here.

We thank the reviewer for the comments. As these suggestions are more thoroughly expressed in the following comments for authors, we will provide the responses in the corresponding sections, as shown below.

Reviewer #2 (Recommendations For The Authors):

Significant concerns:

(1) Establishing the precise localization of MCAK wrt microtubule end is highly non-trivial. More details should be provided, including substantial supplementary data. In particular, the authors claim ~6 nm accuracy in microtubule end positioning - this should be substantiated by data showing individual overlaid microtubule end intensity profiles as well as fits with standard deviations etc. Furthermore, to conclude that MCAK binds behind XMAP215, the authors should look at the localization of the two proteins simultaneously, on the same microtubule end. Notably, EB binding profiles are well known to exponentially decay along the microtubule lattice - this is not very apparent from the presented data. If MCAK's autonomous binding pattern matches that of EB, we should be seeing an exponentially-decaying localization for MCAK as well? However, averaged MCAK signals seem to only be fitted to Gaussian. Note that the EB binding region (i.e. position and size of the EB comet) can be substantially modulated by increasing the microtubule growth rate - this can be easily accomplished by increasing tubulin concentrations or the addition of XMAP215 (e.g. see Maurer et al. Cur Bio 2014). Thus to establish that MCAK on its own binds the same region as EB, experiments that directly modulate the size and the position of this region should be added.

(1) We thank the reviewer for this comment. Regarding the accuracy in microtubule end positioning, we now provide more details, and please see pages 18-19, lines 625-645 in the revised manuscript.

(2) Regarding the relative localization of XMAP215 and MCAK, we performed additional experiments to record their colocalizations simultaneously, on the same microtubule end. Our results showed that MCAK predominantly binds behind XMAP215, with 14.5% appearing within the XMAP215's binding region. Please see Fig. 2.D-E and lines 184-197 in the revised manuscript.

(3) Regarding the exponential decay of the EB1 signal along microtubules, we observed that the position probability distribution measured in the present study follows a Gaussian distribution, and the expected exponential decay was not apparent. Since the exponential decay is thought to result from the time delay between tubulin polymerization and GTP hydrolysis, slower polymerization is expected to reduce this latency (Maurer et al., 2014). In our experiments, the growth rate was relatively low (~0.7 mm/min), much slower than the rate observed in cells, where the comet-shaped EB1 signal is most pronounced. The previous study has shown that the exponential decay of EB1 is more pronounced at growth rates exceeding 3 mm/min in vitro (Maurer et al., 2014). Therefore, we think that the relatively slow growth may account for the observed non-exponential decay distribution of the EB1 signals. The same reason may also explain the distribution of MCAK.

(4) We agree with the reviewer's suggestion that altering microtubule growth rate is a valid and effective approach to regulate the EB cap length. However, the conclusion that MCAK binds to the EB region is supported by three lines of evidence: (1) the localization of MCAK at

the ends of microtubules, (2) new experimental data showing that MCAK binds to the proximal end of the XMAP215 site, and (3) the tendency of MCAK to bind GTPyS microtubules, similar to EB1. Based on these findings, we did not pursue additional experiments to modify the length of the EB cap.

(2) Even if MCAK indeed binds behind XMAP215, there is no evidence that this region is defined by the GDP-Pi nucleotide state; it could still be curved protofilaments. GTPyS is an analogue of GTP - to what extent GTPyS microtubules exactly mimic the GDP-Pi-tubulin state remains controversial. Furthermore, nucleotide sensing for EB is thought to be achieved through its binding at the interface of four tubulin dimers. However MCAK's binding site is distinct, and it has been shown to recognize intradimer tubulin curvature. Thus it is not clear how MCAK would sense the nucleotide state. On the other hand, there is mounting evidence that the morphology of the growing microtubule end can be highly variable, and that curved protofilaments may be protruding off the growing ends for tens of nanometers or more, previously observed both by EM as well as by fluorescence (e.g. McIntosh, Moores, Chretien, Odde, Gardner, Akhmanova, Hancock, Zanich labs). Thus, to establish that MCAK indeed localizes along the closed lattice, EM approaches should be used.

First, we conducted additional experiments that demonstrate MCAK indeed binds behind XMAP215, supporting the conclusion that MCAK interacts with the EB cap (please see Fig. 2 in the revised manuscript). Second, our argument that MCAK preferentially binds to GDP-Pi tubulin is based on two observations: (1) the binding regions of MCAK overlap with those of EB1, and (2) MCAK preferentially binds to GTPyS microtubules, which are considered a close analogue of GDP-Pi tubulin. Third, understanding the structural basis of how MCAK senses the nucleotide state of tubulin is beyond the scope of the present study. However, inspired by the reviewer's suggestion, we looked into the structure of the MCAK-tubulin complex. The L2 loop of MCAK makes direct contact with the interdimer interface (Trofimova et al., 2018; Wang et al., 2017), which could provide a structural basis for recognizing the changes induced by GTP hydrolysis. While this remains a hypothesis, it is certainly a promising direction for future research. Fourth, we agree with the reviewer that an EM approach would be ideal for establishing that MCAK localizes along the closed lattice. However, this is not the focus of the current study. Instead, we argue that MCAK binds to the EB cap, where at least some lateral interactions are likely to have formed.

(3) The physiological relevance of the study is rather questionable: MCAK has been previously established to be able to both diffuse along the microtubule lattice (e.g. Helenius et al.) as well as hitchhike on EBs (Gouveia et al.). Given the established localization of EBs to growing microtubule ends in cells, and apparently higher affinity of MCAK for EB vs. the microtubule end itself (although direct comparisons with the literature have not been reported here), the relevance of MCAK's autonomous binding to dynamic microtubule ends is dubious.

We thank the reviewer for raising the importance of physiological relevance. Please refer to our response to the comment No.1 of reviewer 1. Briefly, we think that the end-binding affinity of MCAK makes a significant contribution for its cellular functions. To elucidate this concept, we now use a simple model shown in Supplementary Appendix-2 (see pages 49-51, lines 1246-1316). In this model, we simplified MCAK and EB1 binding to microtubule ends by considering only these two proteins while neglecting other factors (e.g. XMAP215). Specifically, we considered two scenarios: one in which both proteins freely diffuse in the cytoplasm and another where MCAK is localized to specific cellular structures, such as the centrosome or centromere. Based on the modeling results, we argue that MCAK's functional impact at microtubule ends derives both from its intrinsic end-binding capacity and its ability to strengthen the EB1-mediated end association pathway.

(4) Finally, the study seriously lacks discussion of and comparison with the existing literature on this topic. There are major omissions in citing relevant literature, such as e.g. landmark study by Kinoshita et al. Science 2001. Several findings reported here directly contradict previous findings in the literature. Direct comparison with e.g. Gouveia et al findings, Helenius et al. findings, and others need to be included. For example, Gouveia et al reported that EB is necessary for MCAK plus-end-tracking in vitro (please see Figure 1 of their manuscript). The authors should discuss how they reconcile the differences in their findings when compared to this earlier study.

We thank the reviewer for this helpful suggestion. In the revised manuscript, we have updated the text description and included comparative discussions with other relevant studies in the Discussion section. Specifically, we added comparisons with the research on XMAP215 in page 14, lines 459-472 (Barr and Gergely, 2008; Kinoshita et al., 2001; Tournebize et al., 2000). Additionally, we have compared our findings with those of Gouveia et al. and Helenius et al. regarding MCAK's preference for binding microtubule ends in page 6, lines 145-157 and page 13, 408-441, respectively (Gouveia et al., 2010; Helenius et al., 2006).

Additional specific comments:

Figure 1

Gouveia et al. (Figure 1) reported that MCAK does not autonomously preferentially localize to growing tips. Specifically, Gouveia et al. found equal association rates of MCAK to both the lattice and the tip in the presence of EB3delT, an EB3 construct that does not directly interact with MCAK. How can these findings be reconciled with the results presented here?

We are uncertain why there was no observed difference in the on-rates to the lattice and the end in the study by Gouveia et al. Even when considering only the known affinity of MCAK for curved protofilaments at the distal tip of growing microtubules, we would still expect to observe an end-binding preference. After carefully comparing the experimental conditions, we nevertheless identified some differences. First, we used a 160 nm tip size to calculate the on-rate (k_{on}), whereas Gouveia et al. used a 450 nm tip. Using a longer tip size would naturally lead to a smaller (k_{on}) value. Note that we chose 160 nm for several reasons: (i) a previous cryo-electron tomography study has elucidated that the sheet structures of dynamic microtubule ends have an average length of around 180 nm (Guesdon et al., 2016); (ii) Analysis of fluorescence signals at dynamic microtubule ends has demonstrated that the taper length at the microtubule end is less than 180 nm (Maurer et al., 2014); (iii) in the present study, we estimated that the length of MCAK's end-binding region is approximately 160 nm. Second, in Gouveia et al., single-molecule binding events were recorded in the presence of 75 nM EB3 Δ T, which could potentially create a crowded environment at the tip, reducing MCAK binding. Third, as mentioned in our response to Reviewer 1, we took great care to minimize the interference from purification tags (e.g., His-tag) by ensuring their complete removal during protein preparation. Previous studies reported that retaining the His-tag of MAPs led to a significant increase in binding for microtubules (Maurer et al., 2011; Zhu et al., 2009). We believe that some of the factors mentioned above, or their combined effects, may account for the differences in these two observations.

1C shows the decay of tubulin signal over several hundred nm - should show individual traces? How aligned? Doesn't this long decay suggest protruding protofilaments? (E.g. Odde/Gardner work).

(1) In the revised manuscript, we now show individual traces (e.g. in Fig. 1B and Fig. 2A). The average trace for tubulin signal with standard deviation was shown in Fig. 2C.

(2) The microtubule lattice was considered as a Gaussian wall and its end as a half-Gaussian in every frame. Use the peak position of the half-Gaussian of every frame to align and average microtubule end signals, during the dwell time. The average microtubule ends' half-Gaussian peak used as a reference to measure the intensity profile of individual single-molecule binding event in every frame (see page18, lines 607-624).

(3) We think that the decay of tubulin signal results from the convolution of the tapered end structure and the point spread function. In the revised manuscript, we have updated the Figures to provide unprocessed original data in Fig. 1B and Fig. 2A.

Please show absolute numbers of measurements in 1C (rather than normalized distribution only).

In the revised manuscript, we have included the raw data for both tubulin and MCAK signals as part of the methods description. In Fig. 1, using normalized values allows for the simultaneous representation of microtubule and protein signals on a unified graph.

How do the results in 1D-G compare with the previous literature? Particularly comparison of on-rates between this study and the Gouveia et al? Assuming 1 μm = 1625 dimers, it appears that in the presence of EB3, the on-rate of MCAK to the tips reported in Gouveia et al. is an order of magnitude higher than reported here in the absence of EB3 (4.3×10^{-4} vs. 2×10^{-5}). If so, and given the robust presence of EB proteins at growing microtubule ends in cells, this would invalidate the potential physiological relevance of the current study. Note that the dwell times measured in Gouveia et al. are also longer than those measured here.

Note that in Gouveia et al, the concentration of mCherry-EB3 was 75 nM, about 187.5 times higher than that of MCAK (0.4 nM). The relative concentrations of these two proteins are not always the case in cells. Regarding the physiological relevance of the end-binding affinity of MCAK itself, please refer to our response to the point No.1 of Reviewer 1.

Notably, Helenius et al reported a diffusion constant for MCAK of $0.38 \mu\text{m}^2/\text{s}$, which is more than an order of magnitude higher than reported here. The authors should comment on this!

In the revised manuscript, we have provided an explanation for the difference in diffusion coefficient. Please see page 6, line 142-157. In short, low salt condition facilitates rapid diffusion of MCAK.

Figure 2:

This figure is critical and really depends on the analysis of the tubulin signal. Note significant variability in tubulin signal between presented examples in 2A. Also, while 2C looks qualitatively similar, there appears to be significant variability over the several hundred nm from the tip along the lattice. This is the crucial region; statistical significance testing should be presented. More detailed info, including SDs etc. is necessary.

In the revised manuscript, we have provided raw data in Fig. 1B and Fig. 2A. Additionally, we have provided statistical analysis on the tubulin signals (Fig. 2C) and performed significance test. Please see page 5, lines 111-116 and page 7, lines 179-183 for detailed descriptions.

Insights into the morphology of microtubule ends based on TIRF imaging have been previously gained in the literature, with reports of extended tip structures/protruding

protofilaments (see e.g. Coombes et al. Cur Bio 2013, based on the methods of Demchouk et al. 2011). Such analysis should be performed here as well, if we are to conclude that nucleotide state alone, as opposed to the end morphology, specifies MCAK's tip localization.

We appreciate the reviewer's suggestion and agree that it provides a valid optical microscopy-based approach for estimating microtubule end morphology. However, this method did not establish a direct correlation between microtubule end morphology and tubulin nucleotide status. Therefore, we think that refining the measurement of microtubule end morphology will not necessarily provide more information to the understanding of tubulin nucleotide status at MCAK binding sites. Based on the available data in the present study, there are two main pieces of evidence supporting the idea that MCAK can sense tubulin nucleotide status: (1) the binding regions of MCAK and EB overlap significantly, and (2) MCAK shows a clear preference for binding to GTPγS microtubules, similar to EB1 (we provide a new control to support this, Fig. s4). Of course, we do not consider this to be a perfect set of evidence. As the reviewer has pointed out here and in other suggestions, future work should aim to further distinguish the nucleotide status of tubulin in the dynamic versus non-dynamic regions at the ends of microtubules, and to investigate the structural basis by which MCAK recognizes tubulin nucleotide status.

EB comet profile should be clearly reproduced. MCAK should follow the comet profile.

Please see our 3rd response to the point 1 of this reviewer.

The conclusion that the MCAK binding region is larger than XMAP215 is not firm, based on the data presented. The authors state that 'the binding region of MCAK was longer than that of XMAP215'. What is the exact width of the region of the XMAP215 localization and how much longer is the MCAK end-binding region? Is this statistically significant?

We have revised this part in the revised manuscript (page 6, lines 167-172). The position probability distributions of MCAK and XMAP215 were significantly different (K-S test, $p < 10^{-5}$), and the binding region of MCAK (FWHM=185 nm) was significantly longer than that of XMAP215 (FWHM=123 nm).

MCAK localization with AMPPNP should also be performed here. Even low concentrations of MCAK have been shown to induce microtubule catastrophe/end depolymerization. This will dramatically affect microtubule end morphology, and thus apparent positioning of MCAK at the end.

In the end positioning experiment, we used a low concentration of MCAK (1 nM). Under this condition, microtubule dynamics remained unchanged, and the morphology of the microtubule ends was comparable across different conditions (with EB1, MCAK or XMAP215). Additionally, in the revised manuscript, we present a new experiment in which we recorded the localization of both MCAK and XMAP215 on the same microtubule. The results support the conclusion regarding their relative localization: most MCAK is found at the proximal end of the XMAP215 binding region, while approximately 15% of MCAK is located within the XMAP215 binding region. Please see Fig. 2D-E and page 7, lines 184-197 for the corresponding descriptions.

Figure 3:

For clearer presentation, projections showing two microtubule lattice types on the same image (in e.g. two different colors) should be shown first without MCAK, and then with MCAK.

We thank the reviewer for this suggestion. We have adjusted the figure accordingly. Please see Fig. 4 in the revised manuscript.

Please comment on absolute intensity values - scales seem to be incredibly variable.

The fluorescence value presented here is the result of multiple images being summed. Therefore, the difference in absolute values is influenced not only by the binding affinity of MCAK in different states to microtubules, but also by the number of images used. In this analysis, we are not comparing MCAK in different states, but rather evaluating the binding ability of MCAK in the same state on different types of microtubules.

Given that the authors conclude that MCAK binding mimics that of EB, EB intensity measurements and ratios on different lattice substrates should be performed as a positive control.

We performed additional experiments with EB1, in the revised manuscript, we provide the data as a positive control (please see Fig. s4).

Figure 4:

MCAK-nucleotide dependence of GMPCPP microtubule-end binding has been previously established (see e.g. Helenius et al, others?) - what is new here? Need to discuss the literature. This would be more appropriate as a supplemental figure?

In the present study, we reproduced the GMPCPP microtubule-end binding of MCAK in the AMPPNP state, as shown in several previous reports (Desai et al., 1999; Hertzner et al., 2006). Here, we also quantified the end to lattice binding preference, and our results showed that the nucleotide state-dependence shows the same trend as the binding preference of MCAK to the growing microtubule ends. Therefore, we prefer to keep this figure in the main text (Fig. 5).

Figure 5:

Please note that both MCAK mutants show an additional two orders of magnitude lower microtubule binding on-rates when compared to wt MCAK. This makes the analysis of preferential binding substrate for these mutants dubious.

We agreed with this point. We have rewritten this part. Please see page 10, lines 295-327, in the revised manuscript.

Figure 6:

Combined effects of XMAP215 and XKCM1 (MCAK) have been previously explored in the landmark study by Kinoshita et al. Science 2001, which should be cited and discussed. Also note that Moriwaki et al. JCB 2016 explored the combined effects of XMA215 and MCAK - which should be discussed here and compared to the current results.

We agree with the reviewer. We have revised the discussion on this part. Please see page 11, lines 329-342 and page 14, lines 459-472 in the revised manuscript.

Please report quantification for growth rate and lifetime.

In the revised manuscript, we provide all these data. Please see pages 11-12, lines 343-374.

To obtain any new quantitative information on the combined effects of the two proteins, at the very minimum, the authors should perform a titration in protein concentration.

We agree with the reviewer on this point. In our pilot experiments, we performed titration experiments to determine the appropriate concentrations of MCAK and XMAP215, respectively. We selected 50 nM for XMAP215, as it clearly enhances the growth rate and exhibits a mild promoting effect on catastrophe—two key effects of XMAP215 reported in previous studies (Brouhard et al., 2008; Farmer et al., 2021). Reducing the XMAP215 concentration eliminates the catastrophe-promoting effect, while increasing it would not much enhance the growth rate. For MCAK, we chose 20 nM, as it effectively promotes catastrophe; increasing the concentration beyond this point leads to no microtubule growth, at least in the MCAK-only condition. If there's no microtubule growth, it would be difficult to quantify the parameters of microtubule dynamics, hindering a clear comparison of the combined versus individual effects. Therefore, we think that the concentrations used in this study are appropriate and representative. In the revised manuscript, we make this point clearer (see pages 11 and lines 329-342).

Finally, the writing could be improved for overall clarity.

We thank the reviewer for pointing out this. In the revised manuscript, we conducted a thorough revision and review of the text.

Reviewer #3 (Public Review):

The authors revisit an old question of how MCAK goes to microtubule ends, partially answered by many groups over the years. The authors seem to have omitted the literature on MCAK in the past 10-15 years. The novelty is limited due to what has previously been done on the question. Previous work showed MCAK targets to microtubule plus-ends in cells through association with EB proteins and Kif18b (work from Wordeman, Medema, Walczak, Welburn, Akhmanova) but none of their work is cited.

We thank the reviewer for the suggestion. Some of the referenced work has already been cited in our manuscript, such as studies on the interaction between MCAK and EB1. However, other relevant literature had not been properly cited. In the revised manuscript, we have added further discussion on this topic in the context of existing findings. Please refer to pages 3-4, lines 68-85, and pages 13, lines 425-441.

It is not obvious in the paper that these in vitro studies only reveal microtubule end targeting, rather than plus end targeting. MCAK diffuses on the lattice to both ends and its conformation and association with the lattice and ends has also been addressed by other groups-not cited here. I want to particularly highlight the work from Friel's lab where they identified a CDK phosphomimetic mutant close to helix4 which reduces the end preference of MCAK. This residue is very close to the one mutated in this study and is highly relevant because it is a site that is phosphorylated in vivo. This study and the mutant produced here suggest a charge-based recognition of the end of microtubules.

Here the authors analyze this MCAK recognition of the lattice and microtubule ends, with different nucleotide states of MCAK and in the presence of different nucleotide states for the microtubule lattice. The main conclusion is that MCAK affinity for microtubules varies in the presence of different nucleotides (ATP and analogs) which was partially known already. How different nucleotide states of the microtubule lattice influence MCAK binding is novel. This information will be interesting to researchers working on the mechanism of motors and microtubules. However, there are some issues with some

experiments. In the paper, the authors say they measure MCAK residency of growing end microtubules, but in the kymographs, the microtubules don't appear dynamic - in addition, in Figure 1A, MCAK is at microtubule ends and does not cause depolymerization. I would have expected to see depolymerization of the microtubule after MCAK targeting. The MCAK mutants are not well characterized. Do they still have ATPase activity? Are they folded? Can the authors also highlight T537 and discuss this?

Finally, a few experiments are done with MCAK and XMAP215, after the authors say they have demonstrated the binding sites overlap. The data supporting this statement were not obvious and the conclusions that the effect of the two molecules are additive would argue against competing binding sites. Overall, while there are some interesting quantitative measurements of MCAK on microtubules - in particular in relation to the nucleotide state of the microtubule lattice - the insights into end-recognition are modest and do not address or discuss how it might happen in cells. Often the number of events is not recorded. Histograms with large SEM bars are presented, so it is hard to get a good idea of data distribution and robustness. Figures lack annotations. This compromises therefore their quantifications and conclusions. The discussion was hard to follow and needs streamlining, as well as putting their work in the context of what is known from other groups who produced work on this in the past few years.

We thank the reviewer for the comments. Regarding the physiological relevance of the end-binding of MCAK itself, please refer to our response to the point No.1 of reviewer 1. Moreover, as we feel that other suggestions are more thoroughly expressed in the following comments for authors, we will provide the responses in the corresponding sections, as shown below.

Reviewer #3 (Recommendations For The Authors):

Why, on dynamic microtubules, is MCAK at microtubule plus ends and does not cause a catastrophe?

At this concentration (10 nM MCAK with 16 mM tubulin in Fig. 1; 1 nM MCAK with 12 mM tubulin in Fig. 2), MCAK has little effect on microtubule dynamics in our experiments. Using TIRFM, we were able to observe individual MCAK binding events. Based on these observations, we think that in the current experimental condition, a single binding event of MCAK is insufficient to induce microtubule catastrophe; rather, it likely requires cumulative changes resulting from multiple binding events.

Do the MCAK mutants still have ATPase activity?

The ATPase activities of MCAK^{K525A} and MCAK^{V298S} are both reduced to about 1/3 of the wild-type (Fig. s6).

The intensities of GFP are not all the same on the microtubule lattice (eg 1A). See blue and white arrowheads. The authors could be looking at multiple molecules of GFP-MCAK instead of single dimers. How do they account for this possibility?

In the revised manuscript, we provide the gel filtration result of the purified MCAK, and the position of the peak corresponds to ~220 kDa, demonstrating that the purified MCAK in solution is dimeric (please see Fig.s1 and page 5, lines 101-103). We measured the fluorescence intensity of each binding event. A probability distribution of these intensities was then constructed and fitted with a Gaussian function. A binding event was considered to correspond to a single molecule if its intensity fell within $\mu \pm 2\sigma$ of the distribution. The details of the single-molecule screening process are provided in the revised manuscript (see page 17, lines 574-583).

In addition, we also measured the fluorescence intensity of both MCAK^{sN+M} and MCAK. MCAK^{sN+M} is a monomeric mutant that contains the neck domain and motor domain (Wang et al., 2012). The average intensity of MCAK^{sN+M} is 196 A.U., about 65 % of that of MCAK (300 A.U.), suggesting that MCAK is a dimer (see Fig. s1). Moreover, we think that some of the dim signals may result from stochastic background noise, while others likely represent transient bindings of MCAK. The exposure time in our experiments was approximately 0.05 seconds; if the binding duration were shorter than this, the signal would be lower. It is important to note that in this study, we specifically selected binding events lasting at least 2 consecutive frames, meaning transient binding events were not included. This point has been clarified in the Methods section (see page 17, lines 568-569 and lines 574-583).

Could the authors provide a kymograph of an MT growing, in the presence of MCAK+AMPPNP? Can MCAK track the cap?

Under single-molecule conditions, we observed a single MCAK molecule briefly binding to the end of the microtubule. However, we did not record if MCAK at high concentrations could track microtubule ends under AMPPNP conditions.

In the experiments in Figure 6, the authors should also show the localization of MCAK and XMAP215 at microtubule plus ends in their kymographs to show the two molecules overlap.

Regarding the relative localization of XMAP215 and MCAK, we conducted additional experiments to record their colocalization simultaneously at the same microtubule end. Our results show that MCAK predominantly binds behind XMAP215, with 14.5% of MCAK binding within the XMAP215 binding region. Please see Fig. 2.D-E and page 7, lines 184-197 in the revised manuscript. However, we argue that the effects of XMAP215 and MCAK are additive, and their binding sites do not necessarily need to overlap for these effects to occur.

The authors do not report what statistical tests are done in their graphs, and one concern is over error propagation of their data. Instead of bar graphs, showing the data points would be helpful.

We have now shown all data points in the revised manuscript.

MCAK+AMPPNP accumulates at microtubule ends. Appropriate quotes from previous work should be provided.

We have made the revisions accordingly. Please see page 9, lines 273-276.

Controls are missing. An SEC profile for all purified proteins should be presented. Also, the authors need to explain if they report the dimeric or monomeric concentration of MCAK, XMAP215, etc...

We have provided the gel filtration result for all purified proteins in the revised manuscript (Fig.s1). Moreover, we now make it clear that the concentrations of MCAK and EB1 are monomeric concentration. Please see the legend for Fig. 1, line 893 in the revised manuscript.

Figure 1: the microtubules don't look dynamic at all. This is also why the authors can record MCAK at microtubule ends, because their structure is not changing.

The microtubules are dynamic, but they may appear non-dynamic due to the relatively slow growth rate and the high frame rate at which we are recording. We propose that individual

binding events of MCAK induce structural changes at the nanoscopic or molecular scale, which are not detectable using TIRFM.

I recommend the authors measure the K_{on} and K_{off} for single GFP-MCAK mutant molecules and provide the information alongside their normalized and averaged binding intensities of GFP-MCAK in Fig 5. Showing data points instead of bar graphs would be better.

(1) We measured k_{on} and dwell time for mutants at growing microtubule end. However, we did not perform single-molecule tracking for MCAK's binding on stabilized microtubules. This is mainly because the superimposed signal on the stable microtubule already indicates the changes in the mutant's binding affinity to different microtubule structures, and moreover, the binding of the mutants is highly transient, making accurate single-molecule tracking and calculations difficult.

(2) In the revised figure, we have included the data points in all plots.

When discussing how Kinesin-13 interacts with the lattice, the authors should quote the papers that report the organization of full-length Kinesin-13 on tubulin heterodimers: Trofimova et al, 2018; McHugh et al 2019; Benoit et al, 2018. It would reinforce their model and account for the full-length protein, rather than just the motor domain.

We thank the suggestion for the reviewer. In our manuscript, we have cited papers on full-length Kinesin-13 to discuss the interaction between MCAK and microtubule end-curved structure. Additionally, we have utilized the MCAK-tubulin crystal structure (PDB ID: 5MIO) in Fig. 6, as it depicts a human MCAK, which is consistent with the protein used in our study. This structure illustrates the interaction sites between MCAK and tubulin dimer, guiding our mutation studies on specific residues. Thus, we prefer to use the structure (PDB ID: 5MIO) in Fig.6.

Figure 5A. What type of model is this? A PDB code is mentioned. Is this from an X-ray structure? If so, mention it.

We have now included the structural information in the Figure legend (see page 37, lines 1045).

Figure 5B. It is not possible to distinguish the different microtubule lattices (GTPyS, GDP, and GMPCPP). The experiment needs to be better labelled.

We thank the reviewer for this comment. We have now rearranged the figure for better clarity (see Fig. 6).

"Figure 5D: what are the statistical tests? I don't understand " The statistical comparisons were made versus the corresponding value of 848 GFP-MCAK".

We have made this point clearer in the revised manuscript (see pages 38, line 1078-1080).

What is the "EB cap"? This needs explaining.

We provide this explanation for this, please see page 4, lines 87-89 in the revised manuscript.

Work from Friel and co-workers showed MCAK T537E did not have depolymerizing activity and a reduced affinity for microtubule ends. The work of the authors should be discussed with respect to this previously published work.

We thank the reviewer for this suggestion. In the revised manuscript, we have added discussions on this (see page 10, lines 303-307).

| *The concentration of protein used in the assays is not always described.*

We have checked throughout the manuscript and made revisions accordingly.

| *"Having revealed the novel binding sites of MCAK in dynamic microtubule ends " should be on "we wondered how MCAK may work ..with EB1". This is not addressed so should be removed. Instead, they can quote the work from Akhmanova's lab. Realistically this section should be rephrased as there are other plus-end targeting molecules that compete with MCAK, not just XMAP215 and EB1.*

We have rephrased this section as suggested by this reviewer to be more specific. Please see page 11, lines 329-342.

| *What is AMPCPP?*

It should be "AMPPNP"

| *Typos in Figure 5.*

Corrected

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